

Learning-Optimal Design of Data Markets^{*}

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Abstract

We study the design of data markets with strategic role selection. A continuum of agents each receives a private signal regarding an uncertain state, and decides whether to become a *seller* who trades off their signal for monetary revenue, a *buyer* who receives a signal from sellers and subsequently acts based on two signals, or a *non-participant* who acts solely according to their own signal. Each agent prefers a state-matching action. A market designer commits to a transfer function and aims to optimize information aggregation. For state-independent markets where transfers only rely on an agent's role and the overall measures of buyers and sellers, e.g. the standardized data trading platforms, we show the robust existence of a “digital market for lemons” even at optimum: the least informed agents sell, the moderately informed buy, while the most informed opt out. Nevertheless, a state-dependent market which links the state to sellers' revenue, e.g. blockchain-empowered networks offering data-quality-based rewards, may reverse the role distribution and restore information efficiency. Under the optimal design, the most informed agent(s) sell their signal to all others, whose aggregate action achieves the maximum expected payoff among all markets with a single transaction. Moreover, while the designer may need to pay a net subsidy for implementing a learning-optimal state-independent market, she always makes a non-negative profit from a state-dependent one.

Keywords: data markets, platforms, adverse selection, information aggregation

JEL Classification: D47, D82, D83

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1 Introduction

Data markets have become a visible institutional feature as well as a necessary outcome of the ever-growing digital economy. Typically emerging as platforms such as data exchanges (e.g. Amazon’s AWS Data Exchange) and blockchain-empowered networks (e.g. the Openledger community), they curate and deliver data products, match sellers and buyers, and facilitate transactions of various scales. Several essential characteristics distinguish these markets from traditional marketplaces. First, unlike tangible products, data is non-exclusive, nearly costless to replicate, and always associated with information asymmetry.¹ Second, the roles of market participants are often endogenously decided, i.e. individuals and institutions enter with their own endowment of data, and may strategically choose whether to become buyers or sellers. Third, transactions and transfers take diverse forms. While more conventional markets with standardized products and pricing still persist, the advance of distributed ledgers and smart contracts has enabled novel transfer rules which offer quality-based rewards to data providers.

The ecology of data transactions in China offers rich and intriguing illustrations of the above features of data markets. According to the National Data Administration (2025), with the presence of 10 state-owned data centers and over 40 data exchanges nationwide, there also exist more than 17,000 companies and corporate groups that specialize in customized data products and services. Notably, the majority of data transactions seems to be taking place on the more autonomous and less standardized territories. In 2022, the volume of on-exchange traded data in China only accounted for 5% of the overall market size; the number is expected to reach 20% in 2030 (Shanghai Data Exchange, 2024).

While data markets exhibit an explosive surge in both number and variety, a significant proportion of them suffers from unexpectedly low transaction volume and unstable data quality.² Observations from institutional practices have suggested the necessity of policy intervention, but *whether* and *how* may regulators effectively restore market efficiency remain uncertain (Chen, 2022). To understand the relevant challenges and insights, particularly what inspirations they may bring for market designers, we build a canonical model that captures the above features, and study how a designer optimally configures the market via a transfer rule. By interpreting data as informative signals, we focus on the identification of learning-optimal designs, i.e. the transfer rule that optimizes information aggregation. Our main contribution is to highlight the importance of state dependence in design. On the one hand, even the optimal state-independent market exhibits adverse selection: when transfers cannot rely on the

¹As suggested by the phrase “data dilemma”, while the quality of data for model training and forecasts essentially determines its value, the data becomes valueless to a market participant once its quality is credibly conveyed, i.e. the data gets exposed, before any transaction.

²See Koutroumpis et al. (2020) which documents a rich set of empirical studies.

state, only the least informative agents supply their data to the market. In addition, the optimal design may require the designer to pay a sizable net subsidy. On the other hand, the optimal state-dependent market incentivizes the most informative agents to become suppliers and brings information aggregation to its maximum. It also relieves the pressure on the designer’s budget, in the sense that the designer can always implement an optimal design with a negligible subsidy. In some circumstances, the designer can even make a profit by imposing a uniform, incentive-compatible tax on all agents.

In our model, a designer and a continuum of agents interact over four periods ($t = 0, 1, 2, 3$). At $t = 0$, nature draws a binary state unknown to both the designer and the agents, and each agent observes a private signal according to exogenous state-dependent distributions. At $t = 1$, the designer commits to a transfer scheme governing data transactions: the monetary value each agent receives or pays may depend on both exogenous and endogenous variables in the model. A market is referred to as *state-independent* if the transfers are solely determined by the aggregate measures of buyers and sellers. In contrast, it is *state-dependent* if the transfers may also rely on the realized state and agents’ final actions, as specified below.

At $t = 2$, the agents simultaneously choose one among three possible roles in the data market—seller, buyer, or non-participant. Data transactions ensue such that each buyer receives one randomly selected signal from the sellers; equivalently, each seller sends their signal to a number of buyers equal to the buyer-seller ratio. Then the agents make or receive their transfer payment according to the designer’s scheme. At $t = 3$, the non-sellers (buyers and non-participants) each take a final action based on all available information, including their own signal and acquired data if any. They can either make a costly binary investment that yields a positive profit if and only if it matches the state, or simply abstain with no cost.

It is worth noting that our setup digresses from the classical information design framework (e.g. Bergemann et al. (2018)), in which a single data provider designs a menu of experiments or precision levels for potential buyers. In our model, a private signal represent each agent’s *pre-existing informational precision* about the investment—accumulated sectoral expertise, prior research, or industry experience that the agent already possesses—not an experiment whose informativeness may be manipulated. Self-selected data buyers are therefore not blank-slate agents who acquire data based on identical priors; they carry heterogeneously interim beliefs from their endowed signal, and purchase extra data (i.e. an extra signal) only when doing so is expected to improve the accuracy of the subsequent action. This selectivity is consistent with an empirical feature of data-trading platforms: many listed products attract zero transactions, pointing to buyers who are discerning rather than uninformed.

The designer in our framework seeks to maximize the agents’ aggregate net payoff from investment. We refer to the resulting optimal transfer scheme as a *learning-optimal* design,

because learning from acquiring peers’ signals is the only way to improve the correctness of an agent’s action. Our main results are two-fold as follows.

We first focus on state-independent markets where transfers cannot depend on the realized state. In practice, they correspond to institutionalized data exchanges where data is packaged and priced as a standardized digital commodity. We begin by reducing the complexity of the designer’s problem. Specifically, we show that every design can be equivalently implemented by a simple, two-dimensional instrument: a unit price of data, and a fee or subsidy imposed on sellers. Then we prove the robust emergence of a “digital market for lemons”. That is, given an arbitrary design and in every equilibrium, sellers are less informed than buyers, while buyers are less informed than (at least a significant fraction of) non-participants.

To understand the persisting inefficiency in state-independent markets, note that an agent’s perceived marginal value of acquiring an additional signal from a given distribution—the signal distribution of sellers—is *not* monotone in informativeness of the agent’s own signal. For relatively informed agents who intend to make a costly investment to match the state, an extra signal becomes less valuable as their own signal gets more precise. A direct consequence is that the most informed agents would rather not pay for insignificant informational benefit, analogous to the classical adverse selection. However, for relatively uninformed agents who favor costless abstention, an extra signal only turns worthwhile when their own signal is sufficiently precise so that the extra signal may incentivize them to switch to a costly investment. Therefore among the agents who do participate in the market, strategic self-selection must leave the least informed agents as sellers.

In particular, we characterize the learning-optimal state-independent market, with either (1) zero price and a positive subsidy to sellers, or (2) a positive price, relatively more informed sellers, but a narrowed set of buyers. The digital market for lemon continues as a qualitative equilibrium feature. This design also raises an issue of budget balancing, as it requires a potentially deep pocket from the designer for the subsidy. Moreover, implementation of the designer’s preferred equilibrium hinges crucially on the agents’ indifference breaking rule.

Next, we turn to state-dependent markets which allow the designer to base transfer payments on the realized state and the final actions, or more essentially, the matching rate between them. In reality, these markets are a growing section of the digital economy (e.g. Openledger, AIX), typically empowered by distributed ledger technology and smart contracts to offer rewards based on the quality of shared data. We show that the learning-optimal state-dependent market always achieves efficiency under the constraint of one data transaction; that is, the most informative signal is shared with almost the entire population of agents. As one particular construction, the designer pays each seller i a fixed lump-sum reward, triggered when a sufficiently large measure of the buyers receiving i ’s data invest correctly ex post—that is,

when i 's signal clears a correct-action gate in the realized state. With the reward appropriately calibrated, more-informed agents find themselves better off supplying their data and collecting this reward than acquiring data or investing without participating. The digital market for lemons thus vanishes due to role inversion: the most informed agents are now data sellers while the rest buyers. Better still, the designer pays virtually nothing for implementing the optimal design, as the measure of sellers becomes arbitrarily close to zero. When agents have bounded private beliefs and investment is sufficiently attractive, the designer can even make a positive profit by taxing buyers and sellers uniformly.

Finally, we consider model variations that are relevant for practice and/or interesting for analysis. Our results remain robust in these environments, e.g. when data transactions occur for multiple rounds, or when each agent may assume more than one market role.

Literature Review. This paper contributes to the literature on *markets for information*. A key challenge is to understand what governs a buyer's willingness to pay for additional information. Following Bergemann et al. (2018), we view data as information that refines an agent's decision problem under uncertainty. A buyer's value of learning therefore depends sharply on its initial informational endowment—an insight that traces back to the classical comparison-of-experiments framework of Blackwell (1951) and Blackwell (1953), which provides the foundational ordering of information structures. Early work on selling information in financial markets (Admati and Pfleiderer, 1986, 1990) showed that a signal's value depends on the number of acquirers. For a broader introduction, see Bergemann and Bonatti (2019). Data markets typically feature many potential buyers and sellers, so prices must be consistent with competitive participation on both sides. In contrast to models with a monopolistic data provider (Bergemann and Bonatti, 2015; Bergemann et al., 2018), our environment emphasizes competitive dynamics where agents have limited market power and behave as price takers. This perspective is shared by Acemoglu et al. (2022) who study how data externalities generate pricing inefficiencies, by Jones and Tonetti (2020) who develop a macroeconomic framework for data as a nonrival good, and by Veldkamp (2023) who provides a framework for valuing data as an asset, formalizing the tension between nonrivalry and excludability that also underlies our market design problem. A related policy concern is that platforms acting as data intermediaries may strategically distort data provision in response to regulation: Fang and Kim (2021) show that mandating nondiscriminatory data access can reduce a platform's incentive to supply data, so that the equilibrium quantity of information available in the market falls despite the pro-competitive intent of the regulation. Our model shares the insight that the *terms of data exchange* endogenously govern how much information is produced and traded, though we study this through the lens of transfer-scheme design in a competitive market.

Our work also connects to the literature on *information design*, which studies how an informed principal can strategically disclose information to influence receivers’ actions (Kamenica and Gentzkow, 2011; Bergemann et al., 2017; Bergemann and Morris, 2019). When multiple designers compete to persuade a common audience, the equilibrium information structure can differ markedly from the single-sender benchmark (Gentzkow and Kamenica, 2017); our many-seller market can be viewed as a decentralized analogue of such competition. The equilibrium concept linking information structures to outcome distributions is formalized by Bergemann and Morris (2016), while the duality methods for characterizing optimal disclosure are developed by Dworzak and Martini (2019). A growing literature strand integrates information design with mechanism design. Bergemann et al. (2026) develop an integrated framework showing that both information design and mechanism design problems reduce to maximizing linear functionals subject to majorization constraints. Our paper provides a concrete illustration of this interaction: the designer’s choice of transfer scheme (state-independent vs. state-dependent) endogenously shapes the information structure that emerges in equilibrium—determining who trades, what gets disclosed, and how receivers update their beliefs. This endogeneity is also studied by Éő and Szentes (2007) and Bergemann and Välimäki (2002).

The mathematical foundation of our model draws on the *social learning* literature, which studies how agents update beliefs by observing the actions or signals of others (Acemoglu et al., 2011; Banerjee, 1992; Bikhchandani et al., 1992; Smith and Sørensen, 2000). In canonical models, agents receive private signals and observe predecessors’ choices, leading to information aggregation or cascades. When the environment is non-stationary, the dynamics of learning become substantially richer (Moscarini et al., 2006). While our framework shares the fundamental structure of Bayesian updating from possibly multiple sources, our data market features simultaneous role selection and pooled signal bundles, creating a distinct strategic environment where the *composition* of the seller pool—rather than the *sequence* of actions—determines learning outcomes.

Roadmap. The remainder of the paper is organized as follows. Section 2 introduces the model. Section 3 characterizes equilibria in state-independent markets and identifies the learning-optimal state-independent design. Section 4 characterizes the learning-optimal state-dependent market and shows that it optimizes information aggregation. Section 5 discusses some potentially interesting model variations. Section 6 concludes. Appendix A contains all proofs of the main results. Appendix B presents a numerical illustration of the main theoretical findings, as well as additional results on equilibrium selection and comparative statics.

2 Model

Players and timeline. One designer (she) and a continuum of agents (he) with unit measure interact sequentially on a discrete timeline consisting of four stages: $t = 0, 1, 2, 3$.

Agents' information structure. There is a binary state $\omega \in \{l, r\} \equiv \Omega$, with common prior $\Pr(\omega = r) = \Pr(\omega = l) = \frac{1}{2}$ among the designer and the agents. Each agent i observes a private signal $s_i \in (-1, 1) \equiv I$ about ω . Conditional on ω , signals are i.i.d. with state-dependent distribution functions (F_r, F_l) and densities (f_r, f_l) . We impose three initial conditions on these densities:

(i) symmetry: $f_r(s) = f_l(-s)$,

(ii) the *strict monotone likelihood ratio property* (MLRP): for any $s > s'$,

$$\frac{f_r(s)}{f_l(s)} > \frac{f_r(s')}{f_l(s')},$$

(iii) continuity on $(-1, 1)$.

An agent's interim belief becomes more precise as $|s_i|$ increases. Specifically, the belief induced by signal value s is

$$\theta(s) := \Pr(\omega = r \mid s) = \frac{f_r(s)}{f_r(s) + f_l(s)}.$$

Under the above assumptions, $\theta(s)$ is strictly increasing in s , so we can use either s or $\theta(s)$ as the agent's type. For the ease of exposition, we use $\theta(1)$ to denote $\lim_{s \rightarrow 1} \theta(s)$. As a common terminology in theories of social learning, the agents are said to have *unbounded private beliefs* if $\theta(1) = 1$, and *bounded private beliefs* if $\theta(1) < 1$.

Although the display above conditions on $\omega = r$, the symmetry $f_r(s) = f_l(-s)$ implies $\theta(-s) = 1 - \theta(s)$, so $\theta(s)$ is a sufficient statistic for the informativeness of signal s about either state; we therefore also refer to $\theta(s)$ as the *signal precision* of s .

Game. The game proceeds as follows.

Stage 0 (data generation). Nature draws $\omega \in \{l, r\}$. Each agent observes his private signal.

Stage 1 (market design). The designer announces a market transfer rule $\mathcal{T}$, which will be specified below.

Stage 2 (role choice and transaction). Each agent i simultaneously chooses

$$a_i^1 \in \{\mathbf{S}, \mathbf{B}, \mathbf{N}\},$$

where $\mathbf{S}, \mathbf{B}$ and $\mathbf{N}$ refer to *seller*, *buyer* and *non-participant* respectively. We also say that an agent *opts out* when he does not participate in the market. We shall focus on the cases where (1) the set of sellers (denoted I_S), and the set of buyers (denoted I_B) are both Lebesgue measurable; (2) conditional on either state, the signal distribution on I_S , denoted $F_\omega(\cdot|I_S)$, is well-defined whenever $I_S \neq \emptyset$. A buyer then receives an additional signal according to $F_\omega(\cdot|I_S)$. Each seller's signal is thus received, in expectation, by $\frac{|I_B|}{|I_S|}$ buyers whenever $I_S \neq \emptyset$. We allow for $\frac{|I_B|}{|I_S|} = +\infty$ if $I_S \neq \emptyset$ but $|I_S| = 0$.

For a seller i with signal s_i , we use $I_B(s_i)$ to denote the set of buyers who receive s_i . We assume that buyers, in terms of their own signals, are distributed entirely *pro rata* among sellers. That is, each $i \in I_S$ provides his signal s_i to (in expectation) $\frac{|I_B|}{|I_S|}$ buyers, whose private signals are i.i.d. according to $F_\omega(\cdot|I_B)$.

Stage 3 (investment). Each *non-seller*, i.e. agent i with $a_i^1 \neq S$, chooses an investment action,

$$a_i^2 \in \{L, R, Out\}.$$

The action *Out* here means not investing in either direction and differs from the phrase “opting out” in Stage 2, which means non-participation in the data market. For consistency of notations, we let $a_i^2 \equiv \emptyset$ for each seller.³ Let $A_L(s_i), A_R(s_i) \in [0, \frac{|I_B|}{|I_S|}]$ denote the (per-agent) measures of buyers taking actions L and R in $I_B(s_i)$.

Market design. In Stage 2 of the game, the designer (D) commits to a transfer rule that contains two functions, $\mathcal{T} \equiv \{T_S, T_B\}$, where

$$\begin{cases} T_S : \underbrace{\{l, r\}}_{\omega} \times \underbrace{\mathbb{R}_+ \times \mathbb{R}_+}_{A_L(s_i) \text{ and } A_R(s_i)} \times \underbrace{[0, 1] \times [0, 1]}_{|I_S| \text{ and } |I_B|} \rightarrow \mathbb{R} \\ T_B : \underbrace{[0, 1] \times [0, 1]}_{|I_S| \text{ and } |I_B|} \rightarrow \mathbb{R} \end{cases}$$

denote the amount of transfer received by each seller and buyer, respectively, and a non-participant is naturally assumed to receive no transfer. Both T_S and T_B have the measure

³Our baseline model rules out investment by data sellers, which is consistent with the institutional practice. In Section 5.2, we analyze markets allowing data sellers to make investment, and show that our main results are without loss of generality. The literature on information sharing among competitors provides an insightful discussion on possibly confounding incentives when market participants take up dual roles (Novshek and Sonnenschein, 1982; Clarke, 1983; Vives, 1988; Pagano and Jappelli, 1993).

of buyers/sellers as an input variable. In addition, T_S may depend on the realized state and the ex-post measure of either investment action taken.⁴

We shall discuss two categories of markets in the subsequent analysis. If the seller's transfer is invariant with respect to ω and the action measures $(A_L(s_i), A_R(s_i))$, i.e. T_S can be equivalently written as

$$\hat{T}_S : \underbrace{[0, 1] \times [0, 1]}_{|I_S| \text{ and } |I_B|} \rightarrow \mathbb{R},$$

we refer to the market as *state-independent*. Otherwise it is *state-dependent*.

Figure 1 depicts the game and an example of data transaction.

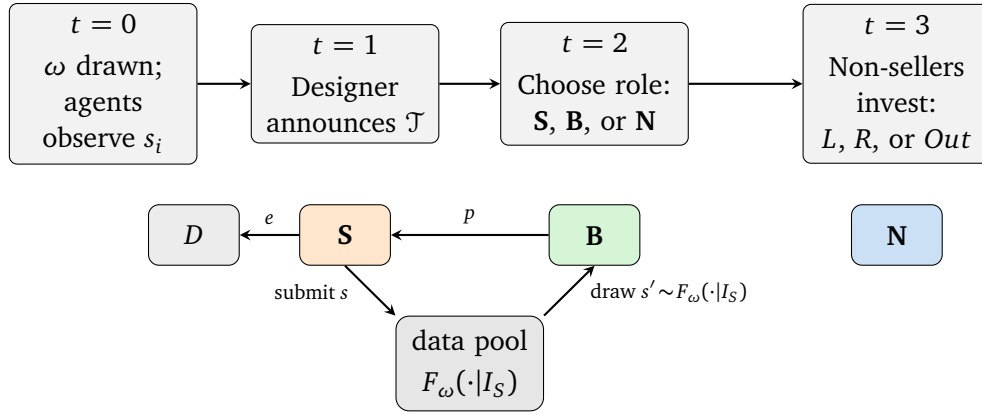


Figure 1: Game timeline (top panel) and role allocation (bottom panel). The bottom panel depicts a simple example for a state-independent market: the designer (D) collect entry fee e from sellers; buyers pay unit price p to sellers. Sellers (S) submit their signals forming a conditional distribution $F_\omega(\cdot|I_S)$, from which buyers (B) each draw one signal at random; non-participants (N) do not receive any additional signal.

Agents' payoffs and equilibrium. An agent's payoff from **Stage 3** depends on the realized state and the investment action: a correct investment yields $V - c$, an incorrect investment yields $-c$, and opting out yields 0, where $V > c > 0$ is common across all agents. Formally, $u_i(\omega, a_i^2)$ is given by

$$u_i(l, L) = u_i(r, R) = V - c, \quad u_i(l, R) = u_i(r, L) = -c, \quad u_i(\omega, Out) = 0.$$

We assume that

$$\theta(1)V > c,$$

⁴Clearly, our results in subsequent sections will hold if T_B is a richer function with more input variables.

i.e. an agent with an extreme signal strictly prefers investment to opting out. Let $U_i(a_i^1, a_i^2, \omega)$ be agent i 's overall payoff from the game. We have

$$\begin{cases} U_i(\mathbf{S}, \emptyset, \omega) = T_S(\omega, A_L(s_i), A_R(s_i), |I_S|, |I_B|) \\ U_i(\mathbf{B}, a_i^2, \omega) = T_B(|I_S|, |I_B|) + u_i(\omega, a_i^2) \\ U_i(\mathbf{N}, a_i^2, \omega) = u_i(\omega, a_i^2). \end{cases} \quad (1)$$

Our solution concept for the agents' strategic behavior, holding $\mathcal{T}$ fixed, is the standard perfect Bayesian equilibrium (PBE) in pure strategies. Without loss of generality, we focus on symmetric equilibria, i.e. $a_i^1(s) = a_i^1(-s) \forall s \in (0, 1)$.

Designer's objective. The designer has deep pocket and is benevolent. That is, it chooses $\mathcal{T}$ to maximize the agents' overall expected net payoff from investment:

$$\mathcal{V}(\mathcal{T}) = \sup_{\mathcal{E}(\mathcal{T})} \left\{ V \times \mathbb{E} \left[\int_i \mathbf{1}\{\omega = l, a_i^2 = L \text{ or } \omega = r, a_i^2 = R\} di \right] - c \times \mathbb{E} \left[\int_i \mathbf{1}\{a_i^2 \neq \text{Out}\} di \right] \right\},$$

where $\mathcal{E}(\mathcal{T})$ is the set of (symmetric) equilibria among agents under $\mathcal{T}$. We assume that for every $\mathcal{T}$, the designer is able to select its preferred equilibrium within $\mathcal{E}(\mathcal{T})$. In the following analysis, we shall simply refer to a designer-preferred equilibrium as an equilibrium.⁵ Since the agents may only improve their probability of correct action by learning from another signal via data transaction, we shall refer to the designer's optimum as a *learning-optimal* design.

3 State-Independent Markets

In this section, we study the learning-optimal design for state-independent markets. Recall that in such a market, each agent's transfer depends solely on their role and the measure of buyers/sellers. As a result and according to (1), the overall expected payoff of each seller is identical, while that of each buyer/non-participant still varies with their private signal.

3.1 Digital Market for Lemons

We proceed in several steps for identifying the designer's optimum. First, we simplify the designer's problem to a great extent, by reducing the design space from the set of arbitrary

⁵This selection device prevents trivial equilibria where no data is traded. In Appendix B, we propose a condition under which there always exists a unique equilibrium.

transfer functions to a particular class of design. Each design in this class is characterized by a two-dimensional parameter, $(p, e) \in \mathbb{R}^2$, where:

- p is the unit data price: each buyer pays p , and total buyer payments are rebated pro rata to sellers, i.e. each seller receives $p \times \frac{|I_B|}{|I_S|}$.
- e is a participation fee collected from sellers (or subsidy if $e < 0$).

It is clear that every (p, e) corresponds to a specific state-independent market. Let $\mathcal{V}'(p, e)$ denote the designer's expected payoff (from her preferred equilibrium) given (p, e) . The following lemmas establish that it is without loss of generality for the designer to focus on such markets with non-negative prices.

Lemma 1 (Problem reduction to price-fee combinations). *For any state-independent transfer scheme $\mathcal{T}$, $\exists(p, e)$ such that $\mathcal{V}'(p, e) \geq \mathcal{V}(\mathcal{T})$.*

Lemma 2 (Optimality of non-negative prices). *In a state-independent market, for any $p < 0$ and $e < 0$ yielding $\mathcal{V}'(p, e)$ for the designer, there exists (p', e') with $p' = 0$ and $e < e' \leq 0$ such that $\mathcal{V}'(p', e') \geq \mathcal{V}'(p, e)$.*

Lemma 1 states that the designer's payoff from every state-independent transfer function can always be weakly improved by some price-fee combination. Intuitively, since a state-independent market implies that the amount of transfer for every buyer (seller) must be identical, the transfer scheme $\mathcal{T}$ can always be implemented by proper selection of (p, e) once the measures of roles $|I_B|$ and $|I_S|$ are given. Specifically, p would equal to the transfer payment by a buyer, and e the difference between two amounts: the transfer payment by a seller, and $p \times \frac{|I_B|}{|I_S|}$. The multiplier $\frac{|I_B|}{|I_S|}$ represents the condition for market clearance where each seller essentially trades with $\frac{|I_B|}{|I_S|}$ buyers in expectation. It then follows that an equilibrium that identifies a particular (I_B, I_S) under $\mathcal{T}$ must imply an equilibrium with identical (I_B, I_S) under (p, e) .

Subsequently, Lemma 2 further constrains the designer's choice space without loss of generality by arguing that negative prices (that is, each buyer receives a subsidy) must be sub-optimal, in the sense that every price-fee combination with a negative price must be dominated by some other combination with zero price. Note that whenever the price is non-positive, a seller must be subsidized and thus $e < 0$. Meanwhile, an agent always weakly prefers buying data to opting out, so it is without loss to focus on equilibria with only buyers and sellers when the designer may ultimately choose her favorable equilibrium. In such an equilibrium, all that matters for an agent's role selection is the difference between expected payoffs from the two roles. Hence for every equilibrium where a buyer is subsidized via a negative price, the designer is always able to induce another equilibrium with identical role choices by canceling the subsidy to buyers (i.e. imposing zero data price) and reducing the subsidy to sellers.

We will henceforth focus on markets characterized by $(p, e) \in \mathbb{R}_+ \times \mathbb{R}$. Our first main result asserts that every such market is a digital market for lemons, where only the least informed agents sell their data.

Theorem 1 (Digital market for lemons). *For arbitrary (p, e) in a state-independent market, whenever an equilibrium induces a positive volume in data trading, i.e. $|I_B| > 0$ and $I_S \neq \emptyset$, there must exist two signal cutoffs $\underline{s}$ and $\bar{s}$, $0 < \underline{s} < \bar{s} \leq 1$, such that*

- (i) *Least informed agents sell or opt out: $a_i^1 \in \{\mathbf{S}, \mathbf{N}\}$ for every agent i with $s_i \in (-\underline{s}, \underline{s})$. In particular, $a_i^1 = \mathbf{S}$ if $p \times \frac{|I_B|}{|I_S|} - e > 0$.*
- (ii) *Moderately informed agents buy: $a_i^1 = \mathbf{B}$ for every agent i with $s_i \in (-\bar{s}, -\underline{s}) \cup (\underline{s}, \bar{s})$.*
- (iii) *Most informed agents buy or opt out: $a_i^1 \in \{\mathbf{B}, \mathbf{N}\}$ for every agent i with $s_i \in (-1, -\bar{s}) \cup (\bar{s}, 1)$. In particular, $a_i^1 = \mathbf{N}$ if $p > 0$.*

To understand the intuition behind a persisting market for lemons, consider the generic case of $p > 0$ and $p \times \frac{|I_B|}{|I_S|} - e > 0$. Note that in every equilibrium, an agent's problem is to select the role that offers the highest expected payoff conditional on his private signal. We may further decompose the problem into several steps. First, a property of monotonicity is clear in the comparison between a seller and a non-participant: as an agent's private signal becomes more precise, he becomes more inclined towards no market participation for a signal-dependent payoff, as compared to selling the data for a fixed payment $(p \times \frac{|I_B|}{|I_S|} - e)$. Thus an agent essentially only faces a binary choice: seller or buyer when he is less informed, and buyer or non-participant when more informed.

As a subsequent observation, a buyer's marginal benefit from an additional signal must be non-monotone in his private signal s_i . When he is less informed, his default final action based solely on s_i would be *Out* which always results in zero payoff. The marginal value of an additional signal then stems from possible encouragement for him to select one of the investment options, *L* or *R*. This value is increasing in $|s_i|$, because as the agent has relatively higher confidence in either state, the set of possible additional signals that may induce investment expands. Conversely, when he is more informed, the default final action becomes investment in the favorable state. An additional signal now would only be helpful when it sufficiently reverts the agent's belief towards the opposite direction, so that a wrong action may be prevented. As $|s_i|$ increases, there are less signal values that could change the agent's mind, and thus the marginal value of an additional signal decreases. Figure 2 illustrates how the expected payoff from each role changes according to private signal value.

The above argument establishes a robust property that in fact holds in every possible equilibrium with data trading: the least informed sell, the moderately informed buy, while the

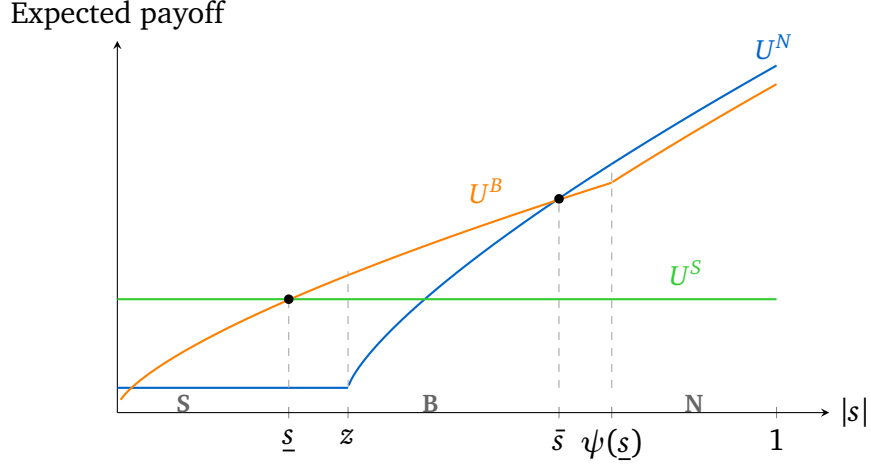


Figure 2: Expected payoffs in equilibrium from each role, assuming $p > 0$ and $e = 0$. On the horizontal axis, z is a signal threshold characterized by $\theta(z) = \frac{c}{v}$, so that a rational non-participant invests if and only if $|s| > z$. The corresponding payoff U^N (blue) is then constant at zero before z and increasing after z . The buyer's expected payoff U^B (orange) rises with diminishing marginal rate, parallel to U^N after the data-relevance frontier $\psi(\underline{s})$, which is the particular signal value beyond which an additional signal from $F(\cdot|I_S)$ provides no value. The seller payoff U^S (green) is signal-independent and hence constant.

most informed opt out. It echoes the classical market for lemons in the sense that owners of the highest-quality assets (most informative signals) stay away from the market. Nevertheless, it also offers a new insight to adverse selection: even if the moderately informed agents do participate in data markets, the value of their signal to information aggregation is still not realized as they self-select as data buyers.

Two special cases arise when p and e take particular values. When $p = 0$, i.e. data is free, the most informed agents are indifferent between buying and opting out, while the free data will not alter their final action in any case. Therefore an arbitrary subset of these agents buying data, with the rest opting out, may occur in equilibrium. When $p \times \frac{|I_B|}{|I_S|} - e = 0$, i.e. sellers earn zero revenue, sellers are indifferent between selling and opting out. Among them, the most informed seller must also be indifferent between buying and selling, implying a non-negative data price. As a result, the two special cases almost never coexist except when the most informed seller would have zero marginal benefit from buying, i.e. $\underline{s}$ is sufficiently low.

Remark: bound of signal thresholds and possibility of market failure. We can easily observe parametric bounds for the signal thresholds $\underline{s}$ and $\bar{s}$. For instance, if $p \times \frac{|I_B|}{|I_S|} - e > 0$, $\underline{s}$ must exceed the signal value such that two identical signals induce indifference between investment and not, otherwise selling always dominates buying. Figure 3 provides an illustration, where x denotes this threshold: letting $\tilde{\theta}(s, s')$ be an agent's belief on $\theta = r$ given two distinct signals

s and s' , then

$$\tilde{\theta}(x, x) = \frac{c}{V}.$$

If $p \times \frac{|I_B|}{|I_S|} - e > 0$, it is possible that $\underline{s} < x$, but as the next subsection will demonstrate, the designer never finds it optimal to implement such an equilibrium and is always able to implement one where $\underline{s} \geq x$. Therefore it is without loss of generality to consider x as a uniform lower bound for $\underline{s}$.

Meanwhile, let

$$\psi(\underline{s}) \equiv \sup\{s \in (0, 1] : \tilde{\theta}(s, -\underline{s}) < \frac{c}{V}\},$$

then it must be the case that $\bar{s} \leq \psi(\underline{s})$. Here $\psi(\underline{s})$ identifies the signal upper bound where additional data is decision-relevant for an agent, i.e. an additional signal from conditional distribution $F(\cdot|I_S)$ may alter his final action. When the agent's private signal goes beyond $\psi(\underline{s})$, he becomes strictly unwilling to acquire data with any positive price, and remains indifferent even if data is free.

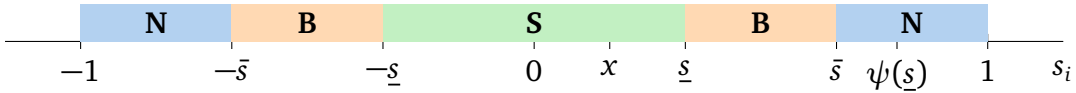


Figure 3: Equilibrium role allocation, assuming $p > 0$ and $e = 0$.

An immediate implication of the above remark is that, when investment is so attractive (low c or high V) that even the least informed agent would like to invest solely based on his signal, there cannot be any equilibrium where data trading actually facilitates information aggregation. In this environment, an agent's marginal benefit from an additional signal is always decreasing in his signal precision. Given the equilibrium pattern identified by Theorem 1, a buyer in equilibrium can only receive a signal less precise than his own, and thus will stick to the same investment decision before and after buying data. Then in every equilibrium, either data trading never occurs, or it does but the acquired signal never matters to the buyer. For the latter case to occur, the designer would have to pay a net subsidy without increasing the accuracy of any agent's final action. In other words, no state-independent design can strictly improve information aggregation. In contrast, an equilibrium with data trading actually hurts the designer because a fraction of agents become buyers and do not invest. The designer therefore achieves her optimum by simply leaving the market alone. The following result summarizes this finding.

Corollary 1 (Total market failure). *When $\frac{c}{V} \leq \frac{1}{2}$, $p = e = 0$ is trivially optimal.*

Discussion: equilibrium uniqueness and comparative statics. Before proceeding to the optimal

state-independent design in the more interesting case of $\frac{c}{v} > \frac{1}{2}$, we briefly discuss the issue of equilibrium selection in our framework. In general, the model may admit multiple equilibria, given which the designer selects her preferred equilibrium for calculating the resulting payoff. In Appendix B, we provide a condition under which the equilibrium among agents must be unique given every (p, e) combination, and show that it is satisfied for a number of typical signal distributions, e.g. a family of exponential distributions. Equilibrium selection would then no longer be a binding requirement for a well-defined designer's problem.

Moreover, equilibrium uniqueness implies clear comparative statics of agents' equilibrium pattern following changes in p or e . A higher data price causes the set of buyers to shrink, in the sense that $\underline{s}$ increases while $\bar{s}$ decreases. Although the average signal precision among sellers improves as more agents, attracted by the higher price, become sellers, the marginal benefit in information is insufficient to compensate for the price increase. Therefore, the previously most informed buyers, who only enjoyed an insignificant net benefit from buying data, now choose not to participate in the market. Meanwhile, a higher fee (lower subsidy) for a seller causes both $\underline{s}$ and $\bar{s}$ to decrease. Heuristically, agents first respond to this incentive change in a way that the previously most informed sellers switch to buyers, simply because they earn less from reduced e . This triggers a subsequent move where the previously most informed buyers opt out, due to reduced informativeness from purchased data.

3.2 Learning-Optimal Design

We now focus on the case of $\frac{c}{v} > \frac{1}{2}$, and characterize the learning-optimal design. For an agent i with signal $s_i > 0$ and $|s| > 0$, we use $MB(s_i, |s|)$ to denote i 's expected marginal benefit from another signal $s' \in \{-|s|, |s|\}$ drawn from distribution (F_r, F_l) .

Proposition 1 (Learning-Optimal design). *Suppose $\frac{c}{v} > \frac{1}{2}$. In a state-independent market, the learning-optimal design (p^*, e^*) , which induces signal thresholds $\underline{s}^*, \bar{s}^*$ such that $\underline{s}^* \in [x, z]$ and $\bar{s}^* \in [z, 1]$, belongs to one of two classes, denoted I and II. In class I:*

$$(i) \quad p^* = 0 \text{ and } e^* = - \int_0^{\underline{s}^*} MB(\underline{s}^*, s) \frac{f_r(s) + f_r(-s)}{F_r(\underline{s}^*) - F_r(-\underline{s}^*)} ds. \text{ The designer pays a total subsidy of}$$

$$|e^*|(F_r(\underline{s}^*) - F_r(-\underline{s}^*)).$$

$$(ii) \quad \bar{s}^* = \psi(\underline{s}^*).$$

$$(iii) \quad \text{Agents with signal } s_i \text{ such that } |s_i| < \underline{s}^* \text{ are sellers, while those with } |s_i| \in (\underline{s}^*, \bar{s}^*) \text{ are buyers.}$$

In class II:

(i) $p^* = MB(\underline{s}^*, \underline{s}^*)$ and $e^* = +\infty$. The designer collects a total revenue of

$$p^*(F_r(\bar{s}^*) - F_r(\underline{s}^*) + F_r(-\underline{s}^*) - F_r(-\bar{s}^*)).$$

(ii) $MB(\bar{s}^*, \underline{s}^*) = MB(\underline{s}^*, \underline{s}^*)$ and $\bar{s}^* \leq \psi(\underline{s}^*)$.

(iii) Agents with signal s_i such that $|s_i| = \underline{s}^*$ are sellers, those with $|s_i| < \underline{s}^*$ or $|s_i| > \bar{s}^*$ opt out, while those with $|s_i| \in (\underline{s}^*, \bar{s}^*)$ are buyers.

In class I, data is always free, while the least-informed agents are still incentivized to become sellers by receiving subsidies. As implied by free data, the set of buyers must include, except the sellers, every agent who finds an additional signal useful, and thus the signal upper bound for buyers coincides with the frontier $\psi(\underline{s}^*)$.

The underlying argument for such a design to be potentially learning-optimal stems from a straightforward improvement principle for the designer. Consider any $(p > 0, e)$ where each seller earns a positive net payoff, and the induced signal thresholds $\underline{s}^*, \bar{s}^*$ in the designers preferred equilibrium. By reducing the data price to zero and adjusting e accordingly downward, the designer can always re-balance the incentives for an agent with the threshold signal s , and thus maintain an identical set of sellers. Meanwhile, the set of buyers must strictly expand, in the sense of a higher s , as data becomes free. Therefore, data must be free at the designer's optimum, while sellers are incentivized by pure subsidies. Buyers, or more precisely, buyers who benefit from data trading, are those whose actions may be altered by an additional signal drawn from $F(\cdot|I_s)$, i.e. agents with $s_i \in [\underline{s}^*, \psi(\underline{s}^*)]$. An agent with signal greater than $\psi(\underline{s}^*)$ may as well receive data for free, but with no marginal benefit whatsoever. Finally, as the above argument establishes a one-to-one relation between the agents' net expected payoff and the buyer-seller signal threshold $\underline{s}^*$, we may define this measure as a continuous function $\tilde{V}(\underline{s}^*)$ and identify $\underline{s}^*$ by solving

$$\max_{\underline{s} \in [x, z]} \tilde{V}(\underline{s}), \quad (2)$$

where x is the feasibility lower bound for $\underline{s}$ defined in the remark following Theorem 1 by $\tilde{\theta}(x, x) = c/V$. We provide a numerical example below.

Example I. For $f_r(s) \propto e^s$, $f_l(s) \propto e^{-s}$ on $(-1, 1)$ with $c/V > 1/2$, the first-order condition $\tilde{V}'(\underline{s}) = 0$ admits a closed-form solution. Setting $\xi = e^{2\underline{s}}$ and $\Lambda = z e^z$ where $z = \frac{1}{2} \ln\left(\frac{c}{V-c}\right)$, the FOC reduces to the cubic $\xi^3 - (3 + 2\Lambda)\xi^2 + (3e^{2z} - 2\Lambda)\xi - e^{2z} = 0$, whose unique root

$\xi^* \in (e^z, e^{2z})$ is given by Cardano's formula (see equation (4) in Appendix B.2), yielding

$$\underline{s}^* = \frac{1}{2} \ln \xi^*, \quad \bar{s}^* = \underline{s}^* + z.$$

In class II, the designer induces a more informative set of sellers, which only contains agents i such that $|s_i| = \underline{s}^*$. Note that the remaining designs to be considered, apart from class I, are those where each seller earns a zero net payoff. By Theorem 1, the set of sellers here must be a subset of $[-\underline{s}, \underline{s}]$ for some $\underline{s} \in [x, z]$, while the most informative seller/least informative buyer, say j , is indifferent between buying and selling. The latter condition characterizes the required data price of such a design, which equals j 's marginal benefit from receiving an additional signal drawn from the seller pool. Given $\underline{s}$, we may then derive the unique design that maximizes the agents' overall expected payoff from their final actions: the agents with exactly $|s_i| = \underline{s}$ sell, and those with $|s_i| < \underline{s}$ opt out. Each (infinitesimal) seller shares his signal with an arbitrarily large number of buyers and thus is required to pay an arbitrarily large fee for maintaining his indifference⁶, while the total fee collected by the designer equals the total payment from buyers. Compared to class I, the increased informativeness of traded signals comes at the cost of a smaller set of buyers, as data bears a positive cost in class II designs.

Similarly, we may identify the optimal class II design by solving a one-variable maximization problem for the designer. Condition (ii) implies that $\bar{s}^*$ can be written as a function of $\underline{s}^*$. For every $s \in (-1, -\underline{s}^*) \cup (\underline{s}^*, 1)$, an agent i with signal $s_i = s$ either receives a signal s' such that $|s'| = \underline{s}^*$ (when $|s| < \bar{s}^*$), or receives no additional signal (when $|s| > \bar{s}^*$). Moreover, as an agent with signal s , $|s| < x$, would never find investment profitable even if he received an additional signal of s , it is never optimal for the designer to implement $\underline{s}^* < x$. Thus the agents' aggregate net payoff can again be written as a continuous function $\hat{V}(\underline{s}^*)$, and the designer can characterize $\underline{s}^*$ by solving

$$\max_{\underline{s} \in [x, z]} \hat{V}(\underline{s}). \quad (3)$$

We provide a numerical example below.

Example II. For the same exponential family, class II corresponds to the degenerate seller-pool limit in which the seller set collapses to $\{\pm \underline{s}^*\}$, so each buyer draws from a two-point distribution weighted by her posterior over ω . Setting $a := e^{2\underline{s}^*}$, $A := a^2$, $\eta := e^{2z} = h/(1-h)$,

⁶An explicit characterization of e^* can be made as follows: (1) for sufficiently small $\epsilon > 0$, let agents with $|s_i| \in (\underline{s} - \epsilon, \underline{s})$ be the sellers; (2) calculate $p > 0$ according to j 's indifference between buying and selling; (3) characterize the set of buyers given p , and calculate e by a seller's zero-payoff condition, and (4) obtain $e \rightarrow \infty$ by letting $\epsilon \rightarrow 0$.

condition (i) yields the closed-form price

$$p^* = \frac{(V - c)(A - \eta)}{(1 + a)^2},$$

and condition (ii) is *linear* in $\xi := e^{2\bar{s}^*}$ (the cubic of Example I drops to first order because the seller pool degenerates), giving

$$\xi^* = \frac{\eta(1 + a + A) - A}{1 + a + A - \eta}, \quad \bar{s}^* = \frac{1}{2} \ln \xi^*.$$

One verifies $\bar{s}^* \in [z, \psi(\underline{s}^*)]$, in line with class II (ii). Unlike Example I, the class II problem for $\underline{s}^*$ does not reduce to a cubic with a useful Cardano-type expression. After substituting the closed-form $\bar{s}^*(\underline{s})$ above, the remaining one-dimensional problem in a is algebraic but of high degree; we solve it numerically and report results across c/V in Appendix B.2.

The final learning-optimal design for state-independent markets can be determined by comparing the designer's payoff from (2) and (3). The design contributes to information aggregation by improving the buyers' accuracy of investment, but it has a number of inevitable drawbacks. A common feature of the two design classes is that they only attenuate but never remove adverse selection, as the more-informed agents (buyers and non-participants) still refrain from sharing their data with others. As for class I, it raises a practical concern of budgeting to the designer, as it can only be implemented by subsidizing the sellers. In contrast, while class II guarantees a positive revenue for the designer, it relies crucially on the agents' indifference breaking rule and the designer's ability to select her preferred equilibrium, in order to induce a relatively informed seller pool that only contains agents with $|s_i| = \underline{s}^*$. In the next section, we show how state-dependent designs enable the designer to fully eliminate these drawbacks and achieve maximum information aggregation.

4 State-Dependent Markets

The key difference made by state-dependent markets is that the transfers to buyers and sellers can now be linked via the realized state. More specifically, when a buyer is rewarded by correct investment (obtaining V), the designer may now exploit the transfer functions to also reward the sellers. Intuitively, an agent with a more precise private signal is now offered an additional incentive to self-select as a seller. It then becomes plausible that, for appropriately designed transfer functions, the less-informed agents buy data from the more-informed.

4.1 Equilibrium with Role Inversion

We shall justify the above intuition with a particular construction of transfer functions.

Imagine the following role selection rule among the agents: for some sufficiently small $\epsilon > 0$, identify a threshold value $s_\epsilon \in (0, 1)$ such that

$$1 - F_r(s_\epsilon) + F_r(-s_\epsilon) = \epsilon.$$

An agent i becomes a buyer if $|s_i| < s_\epsilon$ and a seller if $|s_i| > s_\epsilon$. Clearly, this strategy profile features role inversion between buyers and sellers, in the sense that more-informed agents sell while less-informed agents buy, in contrast to an equilibrium in state-independent markets.

To demonstrate heuristically that this particular role selection can be implemented in a state-dependent market, consider a transfer function characterized by a single parameter $\beta_\epsilon > 0$. For the buyers, data is free: they each receive a random signal from $F_\omega(s | |s| > s_\epsilon)$ without payment or subsidy. Meanwhile, each seller receives a subsidy from the designer, the amount of which depends on the number of buyers who take the correct action after receiving his signal. Recall that in a state-dependent market, the transfer faced by a seller i may depend on both the true state ω , and the number of buyers taking either action after receiving s_i , $(A_L(s_i), A_R(s_i))$.

We set the specific subsidy amount as follows. Imagine a hypothetical seller i with an arbitrary $s_i \in (-1, 1)$. It is clear that if ϵ is sufficiently close to zero, $A_L(s_i)$ is strictly decreasing in s_i while $A_R(s_i)$ is strictly increasing in s_i . If $\omega = l$, the subsidy to seller i is

$$\begin{cases} 0, & \text{if } A_L(s_i) < A_L(-s_\epsilon) \\ \beta_\epsilon, & \text{otherwise.} \end{cases}$$

Likewise, if $\omega = r$, the subsidy to seller i is

$$\begin{cases} 0, & \text{if } A_R(s_i) < A_R(s_\epsilon) \\ \beta_\epsilon, & \text{otherwise.} \end{cases}$$

In practice, one may interpret this transfer policy as linking the seller's revenue to the actual effect of his data in a simple, binary way: he is rewarded a lump-sum payoff β_ϵ if and only if a sufficiently large measure of buyers invest wisely ex-post, thanks to his data. We use $\mathcal{T}(\epsilon)$ to denote a state-dependent market as such, and show that role inversion does occur with appropriately chosen β_ϵ .

Theorem 2 (Role inversion). $\exists \bar{\epsilon} \in (0, 1)$ such that for every $\epsilon \in (0, \bar{\epsilon})$, $\exists \beta_\epsilon \in \left(0, \frac{V-c}{\theta(s_\epsilon)}\right]$ where

$\mathcal{T}(\epsilon)$ induces

$$a_i^1(s_i) = \begin{cases} \mathbf{B}, & \text{if } |s_i| < s_\epsilon \\ \mathbf{S}, & \text{if } |s_i| > s_\epsilon \end{cases}$$

in some equilibrium among agents.

Even though data is costless to obtain as a buyer, $\mathcal{T}_\epsilon$ incentivizes the most informed agents to supply their data to the market instead. This is because this state-dependent design creates a linkage between an agent's certitude of the true state, as reflected by his signal, and his revenue as a seller: a sufficiently informative signal, when indicating the correct state, clears the correct-action gate and earns the lump-sum reward, while a less informative signal most definitely earns nothing. Moreover, as an agent is incentivized to sell as long as the expected reward exceeds the corresponding buyer payoff, and this payoff is uniformly bounded above by $V - c$ while $\theta(s_\epsilon)$ stays bounded away from zero for small ϵ , the designer does not need an unbounded β_ϵ for such economics to work. Figure 4 illustrates the relevant payoffs leading to equilibrium role selection.

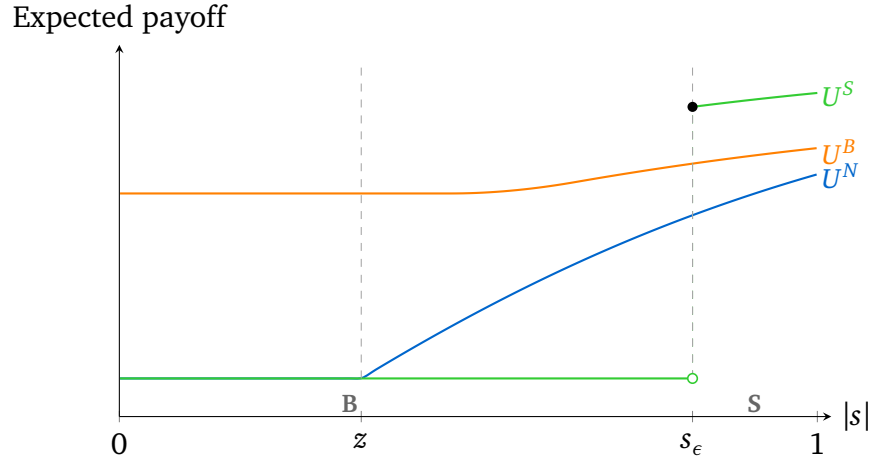


Figure 4: Expected payoffs in equilibrium from each role under $\mathcal{T}_\epsilon$. The seller payoff reflects the gated flat reward: it is zero below s_ϵ and equals $\theta(|s|)\beta_\epsilon$ once the correct-action gate is cleared.

4.2 Learning-Optimal Design and Designer's Profit

As a corollary of Theorem 2, $\mathcal{T}_\epsilon$ enables the designer to approximate the upper bound of information aggregation as $\epsilon \rightarrow 0$, i.e. the sellers' overall signal precision approaches its maximum. We will first explicitly characterize this upper bound.

Consider an agent i with private signal s_i , who receives an additional signal with absolute value $s \in (0, 1)$, such that $s > |s_i|$: he receives signal s (resp. $-s$) with probability $\theta(s)$ and signal $-s$ (resp. s) with probability $1 - \theta(s)$ when $\omega = r$ (resp. $\omega = l$). Let $y(s_i, s)$ denote agent i 's probability of making a correct investment, i.e. $\omega = l$ and $a_i^2 = L$ or $\omega = r$ and $a_i^2 = R$, and let

$$\tilde{s}(s) \equiv \sup\{s_i \geq 0 : \frac{f_l(s_i)f_l(-s)}{f_r(s_i)f_r(-s) + f_l(s_i)f_l(-s)} \geq \frac{c}{V}\}.$$

$\tilde{s}(s)$ identifies the most precise $s_i \geq 0$ such that, after receiving a countervailing signal $-s$, agent i will still invest—but in the state opposite to what s_i indicates—rather than opting out. Respectively, an agent with signal $s_i \in (\tilde{s}, s)$ will not invest in either direction after receiving $-s$. Then the agents' largest possible net payoff from their final action, in a data market where each agent may receive at most one additional signal, can be written as the limit

$$\bar{V} \equiv \lim_{s \rightarrow 1} [\theta(s)(F_r(s) - F_r(-\tilde{s}(s)))(V - c) - (1 - \theta(s))(F_r(\tilde{s}(s)) - F_r(-s))c].$$

$\bar{V}$ can be regarded as an efficiency benchmark of information aggregation, but a constrained one as each agent is still presented with possibly noisy information (2 signals). The following result establishes that $\mathcal{T}_\epsilon$ is an asymptotically learning-optimal design, as it renders a payoff arbitrarily close to $\bar{V}$ for the designer.

Proposition 2 (Efficient information aggregation). $\sup_{\epsilon > 0} \mathcal{V}(\mathcal{T}_\epsilon) = \bar{V}$.

The approximation of $\bar{V}$ stems from the property of $\mathcal{T}_\epsilon$ that as ϵ becomes smaller, only agents with more extreme signals self-select to be sellers. At the limit, i.e. with $\epsilon \rightarrow 0$, the most informed agents (with signals such that $|s| \rightarrow 1$) provide their signal to the entire agent population, who then take the final action accordingly. Thus the agents making a correct investment under $\mathcal{T}_\epsilon$ are those who (1) receive an additional signal indicating the correct true state and (2) form a posterior incentivizing him to invest. Intuitively, for an agent i receiving signal $s > 0$ from some seller under $\mathcal{T}_\epsilon$, (1) occurs with probability $\theta(s)$, while lowest possible s_i satisfying (2) is characterized by $-\tilde{s}(s)$. Therefore $\mathcal{V}(\mathcal{T}_\epsilon)$ coincides with $\bar{V}$ as ϵ becomes arbitrarily small.

Apart from restoring efficiency, a state-dependent market has a second advantage of relieving the pressure on the designer's budget. In $\mathcal{T}_\epsilon$, the gated lump-sum reward is calibrated against the seller types' buying payoff, so β_ϵ is bounded above by $(V - c)/\theta(s_\epsilon)$ and hence remains uniformly bounded for sufficiently small ϵ . Since the measure of sellers converges to zero as $\epsilon \rightarrow 0$, the designer's cost to implement such an equilibrium is negligible.

As a matter of fact, the designer could sometimes make a positive monetary profit from a learning-optimal state-dependent market, which is impossible in a state-independent one.

Formally, given $\epsilon > 0$, we let $\mathbb{T}(\epsilon)$ denote the set of designs $\mathcal{T}$ such that

$$\mathcal{V}(\mathcal{T}) \geq \bar{V} - \epsilon.$$

Proposition 2 already implies that $\mathbb{T}(\epsilon) \neq \emptyset \forall \epsilon > 0$. Let $\Pi(\epsilon)$ be the supremum of the designer's profit, i.e. the net amount of transfers from the agents to the designer, within designs in $\mathbb{T}(\epsilon)$. The following result provides a necessary and sufficient condition for the designer to profit significantly from a learning-optimal design.

Proposition 3 (Designer's profit).

$$\lim_{\epsilon \rightarrow 0} \Pi(\epsilon) \begin{cases} > 0, & \text{if agents have bounded private beliefs and } \frac{c}{V} > \frac{1}{2} \\ = 0, & \text{otherwise.} \end{cases}$$

Proposition 3 states that the designer stands to gain from a learning-optimal design if and only if no agent can be almost certain about the true state from his private signal, and a completely uninformed agent would refrain from investment. Note that in every (asymptotically) learning-optimal design, there cannot be a positive measure of non-participants, while the most informed agents (of measure $\epsilon \rightarrow 0$) share their signal with the others. Thus any positive profit for the designer must result from taxing the buyers, while how much to tax the buyers depends on the minimum payoff difference between a buyer and a non-participant. This minimum is obtained, naturally, as an agent's private signal approaches its upper bound 1 (or lower bound -1), with his acquired signal as a buyer bearing approximately the same precision. If agents have unbounded private beliefs, such an agent is almost certain about the true state given his private signal. He would then be indifferent between buying and not participating, as he would almost surely make a correct investment either way. If $\frac{c}{V} \leq \frac{1}{2}$, such an agent would invest even under a posterior of $\frac{1}{2}$. He would again be indifferent between buying and not participating, because it would be optimal for him to always make the investment indicated by his private signal, even when receiving an additional, opposite-valued signal.

When agents have bounded private beliefs and $\frac{c}{V} > \frac{1}{2}$, the designer can make a profit from, for instance, a simple variation of $\mathcal{T}_\epsilon$. On top of the previous subsidy for sellers, the designer may now charge a uniform fee $\kappa' < \kappa$ for participation in the market as either buyer or seller. The limiting surplus κ is characterized by

$$\theta(1)^2(V - c) - (1 - \theta(1))^2c - \kappa = \theta(1)V - c, \Leftrightarrow \kappa = \theta(1)(1 - \theta(1))(2c - V).$$

κ captures the minimum marginal benefit for a buyer from the additional signal received,

which, under the learning-optimal design, is obtained as the buyer’s own signal approaches its highest precision. This value is strictly positive here because the buyer is not fully convinced about the true state according to his own signal, and would rationally abstain from investment once observing an opposite-valued signal. In equilibrium, every buyer or seller pays κ' willingly because participation in the market increases his payoff by at least κ compared to the outside option of acting solely on his private signal.

Summary: comparing learning-optimal state-independent and state-dependent markets. The following table summarizes several key distinctions between learning-optimal state-independent and state-dependent markets.

Feature\ Optimal design	State-independent	State-dependent
Agent’s role as signal precision $\uparrow$	$S \rightarrow B \rightarrow NP$ (class I) or $NP \rightarrow S \rightarrow B \rightarrow NP$ (class II)	$B \rightarrow S$
Efficient information aggregation	No	Yes
Designer’s monetary profit	negative (class I) or positive (class II)	always non-negative, sometimes positive

Table 1: Comparison between the two learning-optimal markets. We use S (seller), B (buyer) and NP (non-participant) as abbreviations for market roles.

5 Discussion

In this section, we discuss how alternative strategic environments may affect learning patterns and the optimal design. Section 5.1 elaborates on how the quantitative loss in information aggregation, as measured by the difference between the efficiency benchmark and the designer’s optimum, may change in a dynamic environment. Since no such loss occurs in learning-optimal state-dependent markets, we focus on state-independent markets without loss of generality. Section 5.2 considers markets where an agent may simultaneously assume multiple roles.

5.1 Dynamic Data Transactions

To improve information aggregation in state-independent markets, one conceivable way is to consider repeated market participation: if agents can be incentivized to share signals for more than one round, then their overall knowledge about the state can be refined in principle.

Here we consider a simple setting of dynamic markets with a discrete timeline $\tau = 1, 2, \dots$ where

- The realized state remains unchanged over time.
- Both the designer and the agents are myopic.
- At every period τ , the game described in Section 2 is played once, and the designer implements the corresponding learning-optimal state-independent market.
- Each agent remembers all signals that they have received, both initially and via subsequent transactions, over time.

The key variable that changes over time is the signal distribution, or more precisely, the distribution of agents' posterior beliefs after every data transaction. We simulate the dynamic model numerically in Appendix B and find that state-independent markets may not facilitate efficient information aggregation even in a dynamic context. Heuristically, this is not only suggested by numerical results but also holds as a general property.

Two economic forces are at work here. For class-I designs, as the designer is myopic and seeks to maximize the agents' current expected payoff from investment, she becomes unwilling to incentivize more-informed agents, whose measure grows significantly after a few rounds of data transaction, to become sellers. Otherwise, these agents would not make the final investment, which reduces the current base number of investing agents. Hence even though data is free in such designs, the actually traded data contains very limited information. For class-II designs, as the data price remains significantly positive, an agent will always refrain from buying data when he becomes sufficiently informed, which terminates subsequent learning.

As implied by the above argument, a set of sufficient conditions for restoring information efficiency in state-independent markets includes (1) designer's foresight; (2) agents' indifference breaking in favor of buying data and (3) designer's deep pocket for subsidizing data sellers persistently. We shall leave a full-fledged analysis of dynamic data markets for future research.

5.2 Dual Market Roles

One of the key assumptions underlying the outstanding features of our agent equilibrium is that market roles are mutually exclusive. It is conceivable, of course, that in certain scenarios an agent may assume the roles of both buyer and seller, or at least have the choice of doing so with additional costs. We show that our model's main qualitative insights on information aggregation remain robust with this variation.

Assume that in Stage 2 of the game, each agent i chooses

$$a_i^1 \in \{\mathbf{S}, \mathbf{B}, \{\mathbf{S}, \mathbf{B}\}, \mathbf{N}\}.$$

In other words, i may now choose to both sell his data for transfer T_S and buy additional data with transfer T_B , and later in Stage 3 decide on the final investment. We also revise the definition of choice $\mathbf{S}$, in the sense that a seller may also make his investment in Stage 3. Let $\mathbb{T}_I$ be the set of all state-independent designs and $\mathbb{T}_D$ the set of all state-dependent designs. The following result states that similar to 2, information aggregation becomes efficient in learning-optimal state-dependent markets. In state-independent markets, efficient learning is now possible but only under very restrictive conditions.

Proposition 4 (Dual market roles). $\sup_{\mathcal{T} \in \mathbb{T}_I} \mathcal{V}(\mathcal{T}) = \sup_{\mathcal{T} \in \mathbb{T}_D} \mathcal{V}(\mathcal{T}) = \bar{V}$. However, for $\mathcal{T} \in \mathbb{T}_I$ to achieve $\mathcal{V}(\mathcal{T}) \rightarrow \bar{V}$, it is necessary that sellers:

- (i) *earn exactly zero revenue;*
- (ii) *break indifference in a way that only the most informed agents sell.*

For state-dependent markets, the designer may again exploit a variation of $\mathcal{T}_\epsilon$ as her optimum, in order to facilitate efficient learning. Specifically, sellers receive the same gated flat reward as in Theorem 2, and the designer charges a fixed fee from each seller. In this way, less informed agents fail the gate and face an expected loss selling their signals, while sufficiently informed agents clear the gate in the state indicated by their signal; the pattern of role inversion, i.e. only the most informed agents sell, is preserved in equilibrium.

For state-independent markets, first note that whenever the sellers receive a positive transfer, *every* agent must sell their signal.⁷ In this case, information aggregation will never be efficient as there will always be less-informed sellers. The only possibility of efficient learning, therefore, is to implement a non-generic equilibrium as follows. Suppose agents i with signal precision $|s_i| > 1 - \epsilon$, $\epsilon > 0$ sufficiently small, choose to sell. The designer configures p and e so that all agents prefer buying (e.g. p close to zero) but each seller earns zero revenue. The above set of sellers is then incentive compatible, but only weakly by indifference breaking. Compared to the optimal state-dependent market where each agent has a strict preference on role selection, this equilibrium is fragile and difficult to implement in practice.

6 Conclusion

We study a data market where agents strategically choose their roles in data transactions, and we focus on learning-optimal market designs that maximize the agents' overall expected payoff

⁷Suppose that, in a possibly more practical setup, an agent may assume an additional market role by paying an additional cost. For example, he may choose to be a buyer (seller) with no cost, and to also be a seller (buyer) with additional cost $K > 0$. The role allocation of seller $\rightarrow$ buyer $\rightarrow$ non-participant then returns in every agent equilibrium because, as an invariant equilibrium property, the marginal benefit from acquiring an additional signal first increases and then decreases as an agent's private signal becomes more precise.

from investment. Our results explain why undesirable patterns, e.g. low transaction volumes and unstable data quality, may persist in standard data exchanges which are perceived in our framework as state-independent markets. Meanwhile, we predict that state-dependent markets, which establish a linkage between fundamental uncertainties and data sellers' revenue, shall facilitate efficient information aggregation at virtually no cost to the designer.

One can now hardly identify, in their hectic modern life, a product or service that is not at least remotely connected with the utilization of data. The rapid growth of data markets, and digital economy in general, inspires further intriguing questions beyond the scope of this paper. In particular, we believe that the emergence of smart contracts, which lays the practical foundation of our state-dependent markets, will bring profound insights to the study of market design in richer contexts.

References

- Acemoglu, D., Dahleh, M. A., Lobel, I., and Ozdaglar, A. (2011). Bayesian learning in social networks. *The Review of Economic Studies*, 78(4):1201–1236.
- Acemoglu, D., Makhdoumi, A., Malekian, A., and Ozdaglar, A. (2022). Too much data: Prices and inefficiencies in data markets. *American Economic Journal: Microeconomics*, 14(4):218–256.
- Admati, A. R. and Pfleiderer, P. (1986). A monopolistic market for information. *Journal of Economic Theory*, 39(2):400–438.
- Admati, A. R. and Pfleiderer, P. (1990). Direct and indirect sale of information. *Econometrica*, 58(4):901–928.
- Banerjee, A. V. (1992). A simple model of herd behavior. *The Quarterly Journal of Economics*, 107(3):797–817.
- Bergemann, D. and Bonatti, A. (2015). Selling cookies. *American Economic Journal: Microeconomics*, 7(3):259–294.
- Bergemann, D. and Bonatti, A. (2019). Markets for information: An introduction. *Annual Review of Economics*, 11:85–107.
- Bergemann, D., Bonatti, A., and Smolin, A. (2018). The design and price of information. *American economic review*, 108(1):1–48.
- Bergemann, D., Brooks, B., and Morris, S. (2017). First-price auctions with general information structures: Implications for bidding and revenue. *Econometrica*, 85(1):107–143.
- Bergemann, D., Heumann, T., and Morris, S. (2026). Information design and mechanism design: An integrated framework. *arXiv preprint*, 2601(17267).
- Bergemann, D. and Morris, S. (2016). Bayes correlated equilibrium and the comparison of information structures in games. *Theoretical Economics*, 11(2):487–522.
- Bergemann, D. and Morris, S. (2019). Information design: A unified perspective. *Journal of Economic Literature*, 57(1):44–95.
- Bergemann, D. and Välimäki, J. (2002). Information acquisition and efficient mechanism design. *Econometrica*, 70(3):1007–1033.

- Bikhchandani, S., Hirshleifer, D., and Welch, I. (1992). A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5):992–1026.
- Blackwell, D. (1951). Comparison of experiments. In *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, pages 93–102. University of California Press.
- Blackwell, D. (1953). Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272.
- Chen, Q. (2022). China wants to put data to work as an economic resource—but how? *DigiChina by Stanford Cyber Policy Center*. Available at: <https://digichina.stanford.edu/work/china-wants-to-put-data-to-work-as-an-economic-resource-but-how/>.
- Clarke, R. N. (1983). Collusion and the incentives for information sharing. *The Bell Journal of Economics*, 14(2):383–394.
- Dworczak, P. and Martini, G. (2019). The simple economics of optimal persuasion. *Journal of Political Economy*, 127(5):1993–2048.
- Éső, P. and Szentes, B. (2007). Optimal information disclosure in auctions and the handicap auction. *Review of Economic Studies*, 74(3):705–731.
- Fang, H. and Kim, S. J. (2021). Data neutrality and market competition. Working Paper, University of Pennsylvania Center for Technology, Innovation and Competition.
- Gentzkow, M. and Kamenica, E. (2017). Competition in persuasion. *Review of Economic Studies*, 84(1):300–322.
- Jones, C. I. and Tonetti, C. (2020). Nonrivalry and the economics of data. *American Economic Review*, 110(9):2819–58.
- Kamenica, E. and Gentzkow, M. (2011). Bayesian persuasion. *American Economic Review*, 101(6):2590–2615.
- Koutroumpis, P., Leiponen, A., and Thomas, L. D. W. (2020). Markets for data. *Industrial and Corporate Change*, 29(3):645–660.
- Moscarini, G., Ottaviani, M., and Smith, L. (2006). Social learning in a changing world. *Economic Theory*, 28(3):475–507.

- National Data Administration (2025). National data resources survey report (2025). Technical report, National Data Administration of China. Available at: <https://www.nda.gov.cn/sjj/ywpd/sjzy/0429/ff808081-9b5e8626-019d-d86d1ac9-115b.pdf>.
- Novshek, W. and Sonnenschein, H. (1982). Fulfilled expectations cournot duopoly with information acquisition and release. *The Bell Journal of Economics*, 13(1):214–218.
- Pagano, M. and Jappelli, T. (1993). Information sharing in credit markets. *The journal of finance*, 48(5):1693–1718.
- Shanghai Data Exchange (2024). Data trading within exchange platforms represents approximately 5% of total data trading. Official bulletin. Available at: https://www.chinadep.com/bulletin/news/CTC_20240304135321649352.
- Smith, L. and Sørensen, P. (2000). Pathological outcomes of observational learning. *Econometrica*, 68(2):371–398.
- Veldkamp, L. (2023). Valuing data as an asset. *Review of Finance*, 27(5):1545–1562.
- Vives, X. (1988). Aggregation of information in large cournot markets. *Econometrica*, 56(4):851–876.

A Proofs

The proofs follow the order in which they are referenced in the main text.

Notation for posterior beliefs. Throughout Appendix A, we use three distinct notations for posterior beliefs: $\theta(s)$ denotes the posterior belief given one signal s ; $\hat{\theta}$ denotes a realized value for the posterior belief; and $\tilde{\theta}(s, s')$ denotes the posterior belief formed from two distinct signals s and s' .

Notation for designer values. We write $\mathcal{V}(\mathcal{T})$ for the designer's value at her preferred equilibrium under a general state-independent transfer scheme $\mathcal{T}$, and $\mathcal{V}'(p, e)$ for the corresponding value under the canonical mechanism with instruments (p, e) defined by a unit data price p and a seller participation fee e .

A.1 Proof of Lemma 1

Fix a state-independent transfer scheme $\mathcal{T} = \{\hat{T}_S, \hat{T}_B\}$, and let ρ^* be a designer-preferred equilibrium under $\mathcal{T}$, inducing seller set I_S and buyer set I_B . Let $\mathcal{V}(\mathcal{T})$ denote the designer's value at ρ^* . The case $I_S = \emptyset$ is trivial: no trade occurs under $\mathcal{T}$, and $(p, e) = (0, \bar{e})$ with $\bar{e}$ large enough to deter every seller yields the same no-trade value. We henceforth assume $I_S \neq \emptyset$.

By state-independence, at ρ^* every seller receives $\hat{T}_S(|I_S|, |I_B|)$, every buyer receives $\hat{T}_B(|I_S|, |I_B|)$, and every non-participant receives 0. When $|I_S| > 0$, consider the canonical mechanism with instruments

$$p := -\hat{T}_B(|I_S|, |I_B|), \quad e := \frac{|I_B|}{|I_S|} p - \hat{T}_S(|I_S|, |I_B|).$$

The case $|I_S| = 0$ with $I_S \neq \emptyset$ is admitted as the limit of thin continua $[\underline{s} - \varepsilon, \underline{s}]$ as $\varepsilon \rightarrow 0$; formally, $\frac{|I_B|}{|I_S|} \rightarrow +\infty$ and the canonical instruments (p, e) are taken accordingly with $e \rightarrow +\infty$. At measures $(|I_S|, |I_B|)$, this mechanism gives each buyer $-p = \hat{T}_B(|I_S|, |I_B|)$, each seller $\frac{|I_B|}{|I_S|} p - e = \hat{T}_S(|I_S|, |I_B|)$, and each non-participant 0. Thus the per-role monetary transfers at ρ^* coincide under $\mathcal{T}$ and under (p, e) . Since the seller set is also I_S in both cases, buyers face the same additional-signal distribution $F_\omega(\cdot | I_S)$; hence the expected investment payoff conditional on each role is also unchanged.

Because agents are atomless, no single deviation alters $(|I_S|, |I_B|)$ or $F_\omega(\cdot | I_S)$. An agent therefore evaluates any deviation from ρ^* against the same role-contingent payoffs under either mechanism, so every role-choice inequality at ρ^* is preserved and ρ^* remains an equi-

librium under (p, e) , delivering the designer the same value $\mathcal{V}(\mathcal{T})$. The designer's preferred equilibrium under (p, e) can only do (weakly) better, so $\mathcal{V}'(p, e) \geq \mathcal{V}(\mathcal{T})$. $\square$

A.2 Proof of Lemma 2

Fix any measurable seller set I_S with induced signal distribution $F_\omega(\cdot | I_S)$. Define the buyer's gross expected payoff (before transfers) as

$$U^{B,0}(s; I_S) := \mathbb{E}[\pi(\tilde{\theta}(s, s'))], \quad \pi(\hat{\theta}) := \max\{0, \hat{\theta}V - c, (1 - \hat{\theta})V - c\},$$

where $s' \sim F_\omega(\cdot | I_S)$. Let $U^B(s; I_S) := U^{B,0}(s; I_S) - p$ denote the buyer's expected payoff net of transfers when a type- s agent buys one signal drawn from $F_\omega(\cdot | I_S)$, and let $U^S(I_S, I_B)$ be the seller payoff (which is type-independent under state-independence and the canonical implementation). We use $U^N(s)$ as the payoff of non-participants for any type- s agent.

Fix $p < 0$, $e < 0$, and any equilibrium under (p, e) with role sets (I_S, I_B, I_N) . Since $p < 0$ means buyers are paid, every agent weakly prefers buying to opting out, so $I_N = \emptyset$ and $I_B = I \setminus I_S$. We construct (p', e') with $p' = 0$ and $e < e' \leq 0$ replicating this equilibrium.

Set $p' = 0$ and choose e' so that the per-seller net revenue is unchanged:

$$e' = e - \frac{|I_B|}{|I_S|}p.$$

Because $p < 0$ and $e < 0$, we have $e' = e - \frac{|I_B|}{|I_S|}p > e$ (since $-\frac{|I_B|}{|I_S|}p > 0$); and $e' \leq 0$ iff $\frac{|I_B|}{|I_S|}|p| \leq |e|$, which must hold at the original equilibrium (otherwise sellers would earn negative revenue and strictly prefer $\mathbf{N}$, contradicting $I_S \neq \emptyset$). Thus $e < e' \leq 0$.

Seller incentives. By construction, each seller's monetary payoff is identical under (p, e) and (p', e') ; the seller set is also I_S in both cases, so the information environment is unchanged. The atomlessness argument from the proof of Lemma 1 then gives that every seller's role-choice inequalities at this equilibrium are preserved.

Buyer incentives. Under (p', e') , buyers pay nothing, so every agent weakly prefers $\mathbf{B}$ to $\mathbf{N}$; combined with preserved seller incentives, (I_S, I_B) with $I_N = \emptyset$ remains an equilibrium under (p', e') and delivers the designer the same value. Taking the supremum over equilibria yields $\mathcal{V}'(p', e') \geq \mathcal{V}'(p, e)$. $\square$

A.3 Proof of Theorem 1

The proof establishes that the only role assignments consistent with equilibrium are those described in parts (i)–(iii): the characterization is a fixed-point condition specifying which

actions are *valid* at each signal level, not a prediction that a unique action occurs. The argument rests on two monotonicity properties of the payoff functions; we state and prove these as Lemmas 3 and 4 before turning to the main proof.

Lemma 3 (Single-crossing property between buying and selling). *Fix any (p, e) and any measurable seller set I_S (with induced conditional distributions $F_\omega(\cdot | I_S)$ well-defined). Let*

$$\Delta_{BS}(s; I_S) := U^B(s; I_S) - U^S(I_S, I_B).$$

Then $\Delta_{BS}(s; I_S)$ is weakly increasing in $|s|$. Moreover, it is strictly increasing in $|s|$ on any region of types for which buying leads to investment (rather than always choosing Out) with positive probability.

Proof. Under the canonical mechanism (p, e) , the seller payoff is type-independent:

$$U^S(I_S, I_B) = \frac{|I_B|}{|I_S|} p - e.$$

Hence $\Delta_{BS}(s; I_S)$ is increasing in $|s|$ if and only if $U^B(s; I_S)$ is increasing in $|s|$.

Step 1: final action of buyers. After a type- s buyer draws an additional signal $s' \in I_S$, its posterior is

$$\tilde{\theta}(s, s') := \Pr(\omega = r | s, s').$$

Given $\tilde{\theta}(s, s')$, the optimal stage-3 action is

$$a^2(s, s') = \begin{cases} R & \text{if } \tilde{\theta}(s, s') \geq c/V, \\ \text{Out} & \text{if } 1 - c/V < \tilde{\theta}(s, s') < c/V, \\ L & \text{if } \tilde{\theta}(s, s') \leq 1 - c/V. \end{cases}$$

Let $\pi(\hat{\theta}) := \max\{0, \hat{\theta}V - c, (1 - \hat{\theta})V - c\}$ denote the resulting stage-3 payoff as a function of the posterior. Then the buyer's gross expected payoff from buying is

$$U^{B,0}(s; I_S) = \mathbb{E}[\pi(\tilde{\theta}(s, s'))],$$

where the expectation is taken over ω and $s' \sim F_\omega(\cdot | I_S)$. The buyer's net payoff is

$$U^B(s; I_S) = U^{B,0}(s; I_S) - p.$$

Step 2: region notation and error probabilities. For each s , define the regions of seller signals that induce each action:

$$\mathcal{R}(s) := \{s' \in I_S : \tilde{\theta}(s, s') \geq c/V\}, \quad \mathcal{L}(s) := \{s' \in I_S : \tilde{\theta}(s, s') \leq 1-c/V\}, \quad \mathcal{O}(s) := I_S \setminus (\mathcal{R}(s) \cup \mathcal{L}(s)).$$

For $s \geq 0$ (the case $s \leq 0$ is symmetric), the type is initially weakly inclined toward R . Define the (state-contingent) “error” probabilities

$$\alpha(s; I_S) := \Pr(s' \in \mathcal{L}(s) \cup \mathcal{O}(s) \mid \omega = r, s' \in I_S), \quad \beta(s; I_S) := \Pr(s' \in \mathcal{R}(s) \mid \omega = l, s' \in I_S).$$

Thus, under $\omega = r$, with probability $1 - \alpha(s; I_S)$ the buyer ends up choosing R and earns $V - c$, while otherwise it earns 0 (if Out) or $-c$ (if L). Under $\omega = l$, with probability $\beta(s; I_S)$ the buyer chooses R and loses c , while otherwise it earns 0 (if Out) or $V - c$ (if L). Accordingly, the gross payoff can be written in the compact form

$$U^{B,0}(s; I_S) = \theta(s) \left[(1 - \alpha(s; I_S))(V - c) - \alpha_L(s; I_S)c \right] + (1 - \theta(s)) \left[\beta_L(s; I_S)(V - c) - \beta(s; I_S)c \right],$$

where $\alpha_L(s; I_S) := \Pr(s' \in \mathcal{L}(s) \mid \omega = r, s' \in I_S)$ and $\beta_L(s; I_S) := \Pr(s' \in \mathcal{L}(s) \mid \omega = l, s' \in I_S)$. If the model rules out possible switching to L for $s \geq 0$, the terms involving $\mathcal{L}(\cdot)$ drop out and the expression reduces to $\theta(s)(1 - \alpha)(V - c) - (1 - \theta(s))\beta c$.

Step 3: posterior monotonicity in $|s|$. By conditional independence,

$$\frac{\Pr(\omega = r \mid s, s')}{\Pr(\omega = l \mid s, s')} = \frac{f_r(s)}{f_l(s)} \cdot \frac{f_r(s' \mid I_S)}{f_l(s' \mid I_S)}.$$

Under strict MLRP, $f_r(s)/f_l(s)$ is strictly increasing in s , hence for each fixed s' , the posterior $\tilde{\theta}(s, s')$ is strictly increasing in s . By symmetry $f_r(s) = f_l(-s)$, the same implies that $\tilde{\theta}(s, s')$ is strictly increasing in $|s|$.

Step 4: monotonicity of the gross value of buying. Define the value function

$$W(\theta_0; I_S) := \mathbb{E}_{s' \sim F_{\omega}(\cdot \mid I_S)} [\pi(\tilde{\theta}(s'))],$$

where the expectation is over ω and $s' \sim F_{\omega}(\cdot \mid I_S)$, and $\tilde{\theta}(s')$ denotes the posterior belief given initial belief θ_0 and additional signal s' . Then $U^{B,0}(s; I_S) = W(\theta(s); I_S)$.

We establish that $W(\theta_0; I_S)$ is increasing in $|\theta_0 - 1/2|$.

Convexity. For any fixed decision rule $\delta : I_S \rightarrow \{R, L, Out\}$ mapping each acquired signal to

an action, the expected payoff is

$$\theta_0 \mathbb{E}_{s' \sim F_r(\cdot | I_S)}[u_r(\delta(s'))] + (1 - \theta_0) \mathbb{E}_{s' \sim F_l(\cdot | I_S)}[u_l(\delta(s'))],$$

which is affine in θ_0 . Since $W(\theta_0; I_S)$ is the supremum of this expression over all measurable decision rules δ , it is convex in θ_0 as the pointwise supremum of affine functions.

Symmetry. By the symmetry assumption $f_r(s) = f_l(-s)$ and the corresponding symmetry of actions (L and R are symmetric roles), we have $W(\theta_0; I_S) = W(1 - \theta_0; I_S)$ for all $\theta_0 \in [0, 1]$.

Monotonicity in distance from 1/2. Any function W that is convex and symmetric about 1/2 is necessarily increasing in $|\theta_0 - 1/2|$: for $1/2 \leq \theta'_0 < \theta_0$, convexity and symmetry together imply

$$W(\theta_0) \geq \frac{1}{2}W(\theta'_0) + \frac{1}{2}W(2 - \theta'_0) = W(\theta'_0),$$

where the equality uses $W(2 - \theta'_0) = W(\theta'_0)$ and $2 - \theta'_0 > \theta'_0$ lies symmetrically on the other side of 1/2. The same argument applies for $\theta_0 \leq 1/2$.

By Step 3, $\theta(s)$ is strictly increasing in $|s|$, so $|\theta(s) - 1/2|$ is strictly increasing in $|s|$ (on the domain where $\theta(s) \neq 1/2$). Thus $U^{B,0}(s; I_S) = W(\theta(s); I_S)$ is weakly increasing in $|s|$, and hence $U^B(s; I_S) = U^{B,0}(s; I_S) - p$ is also weakly increasing in $|s|$.

Moreover, if buying induces either type s_1 or type s_2 to invest (choose R or L rather than Out) with positive probability, the inequality is strict: the convexity of W is then strict (the experiment is non-trivial and the buyer's optimal action varies on a positive-measure set of realizations), yielding strict monotonicity $U^{B,0}(s_2; I_S) > U^{B,0}(s_1; I_S)$ for $|s_2| > |s_1|$.

Conclusion. Since $U^S(I_S, I_B)$ does not depend on s , the difference $\Delta_{BS}(s; I_S) = U^B(s; I_S) - U^S(I_S, I_B)$ inherits the monotonicity of $U^B(s; I_S)$ in $|s|$ weakly increasing everywhere, and strictly increasing on any region where buying induces investment with positive probability. This completes the proof of Lemma 3. $\square$

Lemma 4 (Single-peakedness of buyer's marginal gain). *Let $z := \theta^{-1}(c/V) \in (0, 1)$. For any fixed I_S , the buyer's net gain from acquiring an additional signal,*

$$\Delta_{BN}(s; I_S) := U^{B,0}(s; I_S) - U^N(s),$$

is single-peaked in $|s|$: weakly increasing on $\{|s| \leq z\}$ and weakly decreasing on $\{|s| \geq z\}$, with the latter strict whenever the additional signal changes the optimal action with positive probability.

Proof. Recall $U^N(s) = \max\{0, \theta(|s|)V - c\}$, which equals 0 on $\{|s| \leq z\}$ and equals $\theta(|s|)V - c > 0$ on $\{|s| > z\}$.

Region $|s| \leq z$. Here $\Delta_{BN}(s; I_S) = U^{B,0}(s; I_S)$, which is weakly increasing in $|s|$ by Lemma 3.

Region $|s| > z$. We establish that Δ_{BN} is weakly decreasing using the convexity-symmetry structure from Lemma 3. Recall that $U^{B,0}(s; I_S) = W(\theta(s); I_S)$ where $W(\theta_0; I_S)$ is convex and symmetric about $1/2$. For $|s| > z$, the agent invests without buying data, earning $U^N(s) = \theta(|s|)V - c$.

The key observation is that for $\theta_0 > c/V$ (i.e., $|s| > z$), the marginal value of moving further from $1/2$ in the convex-symmetric function $W(\theta_0)$ is bounded above by the marginal value of the linear component $\theta_0 V - c$. Specifically, for any $z < s_1 < s_2$ (assuming both positive for concreteness), we have

$$\frac{W(\theta(s_2); I_S) - W(\theta(s_1); I_S)}{\theta(s_2) - \theta(s_1)} \leq V,$$

because W is the upper envelope of affine functions with slopes at most V in absolute value (the payoff from any fixed action grows at rate $\pm V$ in θ_0). Meanwhile,

$$\frac{U^N(s_2) - U^N(s_1)}{\theta(s_2) - \theta(s_1)} = V.$$

Thus $\Delta_{BN}(s_2; I_S) - \Delta_{BN}(s_1; I_S) \leq 0$, establishing weak monotonicity.

For strict monotonicity, suppose the additional signal changes the buyer's optimal action with positive probability. Working in the θ variable, write Δ_{BN} as a function of $\theta_0 := \theta(|s|)$:

$$\Delta_{BN}(\theta_0; I_S) = W(\theta_0; I_S) - (\theta_0 V - c).$$

Since W is convex in θ_0 and $\theta_0 V - c$ is affine in θ_0 , their difference $\Delta_{BN}(\theta_0; I_S)$ is convex in θ_0 . Because $\theta(|s|)$ is strictly increasing in $|s|$, $\Delta_{BN}(s; I_S)$ is also convex in $|s|$ on $\{|s| > z\}$. A convex, weakly decreasing function is either constant or strictly decreasing. When the additional signal influences the optimal action with positive probability, $\Delta_{BN}(z; I_S) > 0$ (the experiment has positive value) yet $\Delta_{BN}(\psi(\underline{s}); I_S) = 0$ (by definition of ψ , data becomes irrelevant). Hence Δ_{BN} is not constant and must be strictly decreasing on $(z, \psi(\underline{s}))$.

These properties are confirmed numerically in Figure 2: along the simulated $U^{B,0}$ and U^N curves, $\Delta_{BN} = U^{B,0} - U^N$ is flat-then-rising on $\{|s| \leq z\}$, attains a peak near $|s| = z$, and is strictly decreasing on $\{|s| > z\}$ (where the curves U^B and U^N converge as $|s| \rightarrow 1$). $\square$

Proof of Theorem 1. Fix any (p, e) and any equilibrium with $|I_B| > 0$ and $I_S \neq \emptyset$.

Part (i): the low end admits selling or opting out. We show that no buyer can have $|s|$ weakly below any seller's, which forces sellers to occupy the lowest- $|s|$ segment.

Suppose for contradiction that some buyer $s' \in I_B$ satisfies $|s'| \leq |s|$ for some seller $s \in I_S$. Their equilibrium conditions require:

- seller s weakly prefers selling: $U^S \geq U^B(s; I_S)$;
- buyer s' weakly prefers buying: $U^B(s'; I_S) \geq U^S$.

By Lemma 3, $U^B(s; I_S) \geq U^B(s'; I_S)$ since $|s| \geq |s'|$. These three inequalities chain to equalities: $U^S = U^B(s; I_S) = U^B(s'; I_S)$. But if $|s| > |s'|$, the strict monotonicity part of Lemma 3 gives $U^B(s; I_S) > U^B(s'; I_S)$ whenever buying leads to investment with positive probability—a contradiction. Hence $|s| = |s'| =: \underline{s}$, meaning the two agents coincide at a common boundary.

Since no seller can have $|s|$ strictly above any buyer's $|s'|$, sellers must form the lowest- $|s|$ segment: every agent with $|s_i| < \underline{s}$ belongs to $I_S \cup I_N$, with $a_i^1 \in \{\mathbf{S}, \mathbf{N}\}$ as the valid actions. The choice between the two depends on the sign of $U^S = \frac{|I_B|}{|I_S|}p - e$: if $U^S > 0$, selling strictly dominates opting out ($U^S > U^N(s) = 0$ for $|s| < z$), so $a_i^1 = \mathbf{S}$ is the unique valid action; if $U^S = 0$, both are equally valid.

Part (ii): the middle is uniquely buying. An agent with $|s| \in (\underline{s}, \bar{s})$ must prefer buying to both alternatives.

Buying over selling: At $\underline{s}$, the boundary agent is indifferent between selling and buying: $U^B(\underline{s}; I_S) = U^S$. By Lemma 3, $U^B(s; I_S)$ is strictly increasing in $|s|$ on any region where buying leads to investment with positive probability. For $|s| > \underline{s}$, this gives $U^B(s; I_S) > U^B(\underline{s}; I_S) = U^S$, so buying strictly dominates selling.

Buying over opting out: At $\bar{s}$, the boundary agent is indifferent between buying and opting out: $U^B(\bar{s}; I_S) = U^N(\bar{s})$. By Lemma 4, $\Delta_{BN}(s; I_S)$ is strictly decreasing in $|s|$ above z (where $|s| \leq \bar{s}$ places us on the decreasing branch of Lemma 4 whenever $\bar{s} > z$). For $|s| < \bar{s}$, $\Delta_{BN}(s; I_S) > \Delta_{BN}(\bar{s}; I_S) \geq 0$, so $U^B(s; I_S) > U^N(s)$, meaning buying strictly dominates opting out.

Together, buying strictly dominates both selling and opting out for all $|s| \in (\underline{s}, \bar{s})$, so $a_i^1 = \mathbf{B}$ is the unique valid action there.

Part (iii): the high end admits buying or opting out. We first bound the buyer region from above using the data-relevance frontier $\psi(\underline{s})$: for $|s| > \psi(\underline{s})$, the seller signal $s' \sim F_\omega(\cdot | I_S)$ cannot alter the agent's optimal investment decision, so $U^{B,0}(s; I_S) = U^N(s)$ and $\Delta_{BN}(s; I_S) = 0$. Hence $U^B(s; I_S) = U^N(s) - p \leq U^N(s)$, with strict inequality when $p > 0$. In particular, when $p > 0$ no agent with $|s| > \psi(\underline{s})$ chooses to buy, so any buyer cutoff $\bar{s}$ satisfies $\bar{s} \leq \psi(\underline{s}) < 1$.

Define $\bar{s} := \sup\{|s_i| : i \in I_B\}$. At $|s| = \bar{s}$, buying and opting out deliver equal payoff, so $a_i^1 \in \{\mathbf{B}, \mathbf{N}\}$ are both valid. When $p > 0$, Lemma 4 gives that Δ_{BN} is strictly decreasing in $|s|$ for

$|s| > \bar{z}$, so for every $|s| > \bar{s}$, $\Delta_{BN}(s; I_S) < \Delta_{BN}(\bar{s}; I_S) = p$, hence $U^B(s; I_S) < U^N(s)$ and **N** is the unique valid action above $\bar{s}$.

When $p = 0$, $U^B(s; I_S) = U^{B,0}(s; I_S) \geq U^N(s)$ for all s (by Jensen's inequality applied to the convex function π together with the martingale property $\mathbb{E}[\tilde{\theta}(s, s')] = \theta(s)$). Hence buying is weakly preferred to opting out throughout, and both **B** and **N** are valid actions for every $|s| \in [\underline{s}, 1)$.

Characterization of the boundary cases. The “in particular” clauses reflect two binary parameters governing the non-generic boundaries. On the seller side (U^S): when $U^S > 0$, selling strictly dominates opting out for all $|s| \leq \underline{s}$, making $a_1^1 = \mathbf{S}$ unique; when $U^S = 0$, opting out becomes equally valid. In the case $U^S = 0$ and $p > 0$, the seller pool reduces to a symmetric pair of signal values $\{-\underline{s}, +\underline{s}\}$ (a set of measure zero), at which agents are exactly indifferent between selling and opting out. The lower bound for $\underline{s}$ in this case is the threshold x defined by $\tilde{\theta}(x, x) = c/V$: an agent at signal x facing a seller pool consisting of precisely the two signal values $\{-x, +x\}$ draws $+x$ (confirming her inclination) with probability proportional to the likelihood ratio, and $-x$ (contradicting) otherwise. The posterior after a contradicting draw equals $1/2$, yielding action *Out* and payoff 0; after a confirming draw, the posterior is $\tilde{\theta}(x, x) = c/V$, yielding payoff $\tilde{\theta}(x, x)V - c = 0$. Hence the gross buying payoff $U^{B,0}(x; \{-x, +x\}) = 0 = U^N(x)$, so data provides no net value at x , and $\underline{s} \geq x$. On the buyer-non-participant side (p): when $p > 0$, $\bar{s}$ is the unique interior solution to $\Delta_{BN}(\bar{s}; I_S) = p$ on the decreasing branch of Lemma 4; when $p = 0$, both buying and opting out are valid for all $|s| \in [\underline{s}, 1)$. $\square$

A.4 Existence of equilibria with positive trade

Theorem 1 characterizes the role partition conditional on $|I_B| > 0$ and $I_S \neq \emptyset$. We construct such an equilibrium whenever $c/V > 1/2$. By Theorem 1 every equilibrium with positive trade is MB-LS, so it suffices to find $\underline{s}$, $\bar{s}$, and (p^*, e^*) satisfying both margin conditions:

$$U^{B,0}(\underline{s}; I_S) - p^* = U^S \quad (\mathbf{S}|\mathbf{B}), \quad U^{B,0}(\bar{s}; I_S) - p^* = U^N(\bar{s}) \quad (\mathbf{B}|\mathbf{N}).$$

Given $\underline{s}$, the $(\mathbf{S}|\mathbf{B})$ condition pins down p^* from the seller side and $(\mathbf{B}|\mathbf{N})$ pins down p^* from the buyer side; the intermediate value theorem finds the $\bar{s}$ at which both values agree. We treat the continuum case first ($I_S = [-\underline{s}, \underline{s}]$, $U^S > 0$), then the two boundary cases.

Payoff notation. Fix any measurable seller set I_S with induced distribution $F_\omega(\cdot | I_S)$. Let $U^{B,0}(s, I_S) = U^B(s, I_S) + p$ denote type s 's gross expected payoff from buying one signal drawn

from $F_\omega(\cdot | I_S)$ and then choosing $a_i^2 \in \{L, R, Out\}$ optimally. Let the non-participant payoff be

$$U^N(s) := \max\{\theta(s)V - c, 0\}.$$

Definition 1 (Data-useful cutoff). *Fix $c/V > 1/2$. A cutoff $\underline{s} \in (0, 1)$ is data-useful if*

$$U^{B,0}(\underline{s}, I_S(\underline{s})) > U^N(\underline{s}).$$

We say $\underline{s}$ is data-useful of Case 1 if additionally $\theta(\underline{s}) \leq c/V$ (equivalently $U^N(\underline{s}) = 0$). In Case 1, $\underline{s} \in (0, z]$ where z is defined by $\theta(z) = c/V$ (so that $U^N(z) = \theta(z)V - c = 0$). Otherwise, it belongs to Case 2.

Data-usefulness is the viability condition for **(S|B)**: a strictly positive gross buying value at $\underline{s}$ ensures $p^* > 0$ satisfying the margin condition exists, keeping the part (ii) buyer region non-empty.

Lemma 5 (Existence of a Case 1 data-useful cutoff). *Suppose $c/V > 1/2$. Then there exists $\underline{s}$ that is data-useful of Case 1.*

Proof. Let $z := \theta^{-1}(c/V) \in (0, 1)$. For any $s \in (0, z]$ we have $U^N(s) = 0$. Consider $g(s) := U^{B,0}(s, I_S(s))$ on $(0, z]$, where $I_S(s) := [-s, s]$. Under the continuity assumption on (f_r, f_l) , g is continuous.

For s sufficiently close to z , the type has $\theta(s)$ just below the investment threshold c/V , so $U^N(s) = 0$ (opting out). Since the value of information is non-negative, it is clear that $U^{B,0}(s, I_S(s)) \geq U^N(s)$. Moreover, the inequality is strict when the signal is informative and $\theta(s)$ lies at a kink of the convex payoff function $\pi(\theta) = \max\{0, \theta V - c, (1 - \theta)V - c\}$. Since $\theta(s) < c/V$ and strict MLRP ensures the additional signal $s' \in I_S(s)$ is non-degenerate, strict Jensen's inequality yields $U^{B,0}(s, I_S(s)) > \pi(\theta(s)) = 0 = U^N(s)$. Hence $g(s) > 0$ for some $s \in (0, z]$. Choosing such an s as $\underline{s}$ gives $\theta(\underline{s}) \leq c/V$ and $U^{B,0}(\underline{s}, I_S(\underline{s})) > 0$, i.e., a Case 1 data-useful cutoff. $\square$

Lemma 6 (Case 1 data-useful implies existence via intermediate value theorem). *Fix $c/V > 1/2$ and suppose there exists a cutoff $\underline{s}$ that is data-useful of Case 1. For each $\bar{s} \in (\underline{s}, 1)$ define*

$$M(\underline{s}, \bar{s}) := \frac{|I_S(\underline{s})|}{|I_S(\underline{s})| + |I_B(\underline{s}, \bar{s})|} U^{B,0}(\underline{s}, I_S), \quad m(\underline{s}, \bar{s}) := U^{B,0}(\bar{s}, I_S) - U^N(\bar{s}),$$

and $\Phi(\bar{s} | \underline{s}) := m(\underline{s}, \bar{s}) - M(\underline{s}, \bar{s})$. Then there exists $\bar{s}^* \in (\underline{s}, 1)$ such that $\Phi(\bar{s}^* | \underline{s}) = 0$. Let $p^* := m(\underline{s}, \bar{s}^*) = M(\underline{s}, \bar{s}^*)$. Then $(\underline{s}, \bar{s}^*, p^*)$ satisfies both margin conditions **S|B** and **B|N** and hence constitutes an MB-LS equilibrium with $p^* > 0$.

Proof. The maps $m(\underline{s}, \bar{s})$ and $M(\underline{s}, \bar{s})$ are continuous in $\bar{s}$. At $\bar{s} \downarrow z$, Case 1 data-usefulness implies

$$m(\underline{s}, z) = U^{B,0}(z, I_S) - 0, \quad M(\underline{s}, z) = \frac{|I_S(\underline{s})|}{|I_S(\underline{s})| + |I_B(\underline{s}, z)|} U^{B,0}(z, I_S),$$

so $\Phi(z | \underline{s}) > 0$ because $|I_B(\underline{s}, z)| > 0$ and hence the scaling factor in M is strictly below one.

As $\bar{s} \uparrow 1$, the marginal buyer becomes arbitrarily informative relative to the seller pool, so the incremental value of drawing from I_S vanishes and $m(\underline{s}, \bar{s}) \rightarrow 0$; we define at $y \in [z, 1)$

$$m(\underline{s}, y) := U^{B,0}(y, I_S) - U^N(y) \equiv 0,$$

with $y > z$ following from Lemma 4. Meanwhile,

$$M(\underline{s}, y) := \frac{|I_S(\underline{s})|}{|I_S(\underline{s})| + |I_B(\underline{s}, y)|} U^{B,0}(\underline{s}, I_S) > 0,$$

so $\Phi(y | \underline{s}) < 0$.

By the intermediate value theorem there exists $\bar{s}^* \in [z, y]$ with $\Phi(\bar{s}^* | \underline{s}) = 0$. Setting $p^* := m(\underline{s}, \bar{s}^*) = M(\underline{s}, \bar{s}^*)$ yields a price that simultaneously satisfies the **S|B** and **B|N** margin conditions, and $p^* > 0$ since both sides are strictly positive at $\bar{s}^*$. $\square$

Lemma 5 provides a data-useful $\underline{s} \in (0, z]$ with $U^S > 0$, and Lemma 6 locates $\bar{s}^*$ and $p^* > 0$ via IVT, completing the continuum-seller construction.

Boundary case: $U^S = 0, p^* > 0$. Fix any $\underline{s} \in (x, z)$, where x is defined by $\tilde{\theta}(x, x) = c/V$. Set $I_S = \{-\underline{s}, +\underline{s}\}$ (a symmetric pair) with $U^S = 0$, implemented by taking $e \rightarrow -\infty$ (an arbitrarily large per-seller subsidy). The (**S|B**) margin gives $p^* = U^{B,0}(\underline{s}; \{-\underline{s}, +\underline{s}\}) > 0$, which holds for $\underline{s} \in (x, z)$ since $\tilde{\theta}(\underline{s}, \underline{s}) > c/V$ at these cutoffs. The (**B|N**) margin is then satisfied by the same intermediate value argument applied to the degenerate seller distribution, yielding a unique $\bar{s}^* > \underline{s}$.

Boundary case: $U^S = 0, p^* = 0$. At $\underline{s} = x$, $\tilde{\theta}(x, x) = c/V$ implies $U^{B,0}(x; \{-x, +x\}) = 0 = U^N(x)$, so the (**S|B**) margin forces $p^* = 0$. At $p^* = 0$, both buying and opting out are valid for all $|s| \geq x$ (since $U^{B,0} \geq U^N$ everywhere by Jensen's inequality), so $\underline{s} = x$ is the unique feasibility lower bound for the degenerate seller case. $\square$

A.5 Proof of Corollary 1

Suppose toward a contradiction that an equilibrium has $p > 0$, $|I_B| > 0$, and $I_S \neq \emptyset$. Consider any buyer type $s \in I_B$. In equilibrium, buying must weakly dominate not participating:

$$U^B(s; I_S) \geq U^N(s).$$

We show this cannot hold when $c/V \leq 1/2$ and $p > 0$.

Recall that $\pi(\hat{\theta}) = \max\{0, \hat{\theta}V - c, (1 - \hat{\theta})V - c\}$. By Theorem 1, in any equilibrium with trade there exists $\underline{s} > 0$ such that every seller signal satisfies $|s'| \leq \underline{s}$ (i.e. $I_S \subseteq [-\underline{s}, \underline{s}]$, whether I_S is a continuum or merely the pair $\{-\underline{s}, +\underline{s}\}$) and every buyer signal satisfies $|s| > \underline{s}$. Consider any buyer $s \in I_B$ and any seller signal $s' \in I_S$. The posterior likelihood ratio satisfies $\ell(s)\ell(s') > 1$, where $\ell := f_r/f_l$ denotes the likelihood ratio: since $|s| > \underline{s} \geq |s'|$ and the distribution is symmetric ($f_r(t) = f_l(-t)$), MLRP gives $\ell(s) > \ell(\underline{s}) \geq \ell(|s'|) > 1/\ell(|s'|) \geq \ell(s')^{-1}$, so $\ell(s)\ell(s') > 1$. Hence $\tilde{\theta}(s, s') > 1/2$ for every $s' \in I_S$.

When $c/V \leq 1/2$, the investment threshold satisfies $c/V \leq 1/2 \leq \tilde{\theta}(s, s')$ for all $s' \in I_S$. Hence the buyer always invests after purchasing data, and $\pi(\tilde{\theta}(s, s')) = \tilde{\theta}(s, s')V - c$ for all realizations. Therefore,

$$U^{B,0}(s; I_S) = \mathbb{E}_{s'}[\pi(\tilde{\theta}(s, s'))] = \mathbb{E}_{s'}[\tilde{\theta}(s, s')]V - c = \theta(s)V - c = U^N(s),$$

where the second-to-last equality uses the martingale property of Bayesian posteriors. Since $U^B(s; I_S) = U^{B,0}(s; I_S) - p = U^N(s) - p < U^N(s)$ when $p > 0$, no buyer type can prefer buying over non-participation. Hence, no equilibrium with positive trade exists when $p > 0$.

It remains to show that $(p, e) = (0, 0)$ is optimal. By Lemma 2, any $(p < 0, e < 0)$ is weakly dominated by some $(0, e')$ with $e' \leq 0$, so it suffices to consider $p = 0$. In any equilibrium with trade at $p = 0$, Theorem 1 ensures sellers are weakly less informative than buyers, so the same martingale argument gives $U^{B,0}(s; I_S) = U^N(s)$ for every buyer type: data never changes any agent's investment decision. Since $c/V \leq 1/2$ implies every non-seller invests ($\theta(s) \geq c/V$ for all $|s| > 0$), each agent diverted to selling forfeits investment, weakly reducing correct actions. No design can therefore improve upon $(p, e) = (0, 0)$. $\square$

A.6 Ex-post Investment Components P_R and P_L

Fix state $\omega = r$. Sellers $s \in [-\underline{s}, \underline{s}]$ do not invest. Each buyer $s \in [\underline{s}, \bar{s}]$ acquires a signal s' drawn from $G(s'; \underline{s}) := \frac{F_r(s') - F_r(-\underline{s})}{F_r(\underline{s}) - F_r(-\underline{s})}$ on $[-\underline{s}, \underline{s}]$, and invests in R iff $s' \geq \sigma_h(s)$, where $\sigma_h(s)$ is defined by $\ell(s)\ell(\sigma_h(s)) = h/(1-h)$ with $h := c/V$ and $\ell := f_r/f_l$. Direct investors $s \in [\bar{s}, 1] \cup [-1, -\bar{s}]$

invest in the direction of their own signal. Aggregating over f_r , and using symmetry $f_r(s) = f_l(-s)$:

$$P_R = \int_{\bar{s}}^1 f_r(s) ds + \int_{\underline{s}}^{\bar{s}} [1 - G(\sigma_h(s); \underline{s})] f_r(s) ds,$$

$$P_L = \int_{-1}^{-\bar{s}} f_r(s) ds + \int_{-\bar{s}}^{-\underline{s}} G(-\sigma_h(-s); \underline{s}) f_r(s) ds.$$

Welfare $W = (V - c)P_R - cP_L$ is state-independent.

A.7 Proof of Proposition 1

We treat the two seller-pool structures separately. In *Class I*, the seller pool is the continuum $I_S(\underline{s}) = F_\omega(s' | s' \in [-\underline{s}, \underline{s}])$ (positive Lebesgue measure $|I_S(\underline{s})| := F_\omega(\underline{s}) - F_\omega(-\underline{s}) > 0$), yielding $U^S > 0$ and data price $p^* = 0$. In *Class II*, the seller pool is the two-point set $I_S^\dagger(\underline{s}) := \{-\underline{s}, +\underline{s}\}$ (zero Lebesgue measure, so $U^S = 0$ for every finite e , forcing $p^* > 0$ to keep sellers indifferent between selling and buying). In both cases the signal s' acquired by a buyer at threshold $\underline{s}$ satisfies $|s'| = \underline{s}$, so the conditional distribution seen by any buyer is determined by $\underline{s}$ alone; this common structure lets both cases share the welfare expression derived in Appendix A.6. The overall learning-optimum is the better of the two class-optimal designs.

Proof of Class I. The proof proceeds in two steps. Step 1 establishes properties (i) and (ii) by an improvement-principle argument. Step 2 reduces the problem to a one-dimensional optimisation.

Step 1 (Class I): $p^* = 0$, $e^* < 0$, and $\bar{s}^* = \psi(\underline{s}^*)$. Consider any candidate design (p, e) with $p > 0$ supporting an MB-LS equilibrium with thresholds $(\underline{s}, \bar{s})$ and a continuum seller pool $I_S(\underline{s}) = [-\underline{s}, \underline{s}]$. The conditional distribution of acquired signals is $G(s'; \underline{s}) := (F_r(s') - F_r(-\underline{s})) / (F_r(\underline{s}) - F_r(-\underline{s}))$ on $[-\underline{s}, \underline{s}]$, and the seller's equilibrium payoff is $U^S = \frac{|I_B(\underline{s}, \bar{s})|}{|I_S(\underline{s})|} p - e > 0$.

Replace (p, e) by $(0, e')$ with

$$e' := -U^{B,0}(\underline{s}; I_S(\underline{s})).$$

Under $(0, e')$, the marginal seller at $\underline{s}$ earns $-e' = U^{B,0}(\underline{s}; I_S(\underline{s}))$ from selling, which equals the payoff $U^{B,0}(\underline{s}; I_S(\underline{s})) - 0$ from buying at the new zero price. Thus the marginal seller's indifference between selling and buying is preserved, so the seller set $I_S(\underline{s})$ remains an equilibrium supply. Because data is now free, the buyer pool expands up to the data-relevance frontier:

every type $|s_i| \in (\underline{s}, \psi(\underline{s}))$ strictly benefits from acquiring an extra signal, while every type $|s_i| \geq \psi(\underline{s})$ is indifferent.

To see that this expansion strictly raises welfare $W = (V - c)P_R - cP_L$, fix $\underline{s}$ and compute by Leibniz rule,

$$\frac{\partial P_R}{\partial \bar{s}} = -G(\sigma_h(\bar{s}); \underline{s}) f_r(\bar{s}), \quad \frac{\partial P_L}{\partial \bar{s}} = -[1 - G(-\sigma_h(\bar{s}); \underline{s})] f_r(-\bar{s}),$$

both negative for $\bar{s} < \psi(\underline{s})$. The net welfare effect $W_{\bar{s}} = (V - c) \partial P_R / \partial \bar{s} - c \partial P_L / \partial \bar{s} > 0$ rearranges (with $h := c/V$) to

$$h [1 - G(-\sigma_h(\bar{s}); \underline{s})] f_r(-\bar{s}) > (1 - h) G(\sigma_h(\bar{s}); \underline{s}) f_r(\bar{s}). \quad (\star)$$

Letting $c_0 := \sigma_h(\bar{s})$ and substituting the Bayes condition $\ell(\bar{s}) = \frac{h}{(1-h)\ell(c_0)}$, $(\star)$ reduces to

$$\ell(c_0) > \frac{\int_{-\underline{s}}^{c_0} \ell(t) f_l(t) dt}{\int_{-\underline{s}}^{c_0} f_l(t) dt} = \mathbb{E}_{f_l}[\ell(t) \mid t \in [-\underline{s}, c_0]],$$

which holds because ℓ is strictly increasing under MLRP and $c_0 > -\underline{s}$. For $\bar{s} \geq \psi(\underline{s})$, the investment cutoff $\sigma_h(\bar{s}) \leq -\underline{s}$ lies outside I_S , data never alters any buyer's decision, and $\partial P_R / \partial \bar{s} = \partial P_L / \partial \bar{s} = 0$. Welfare is therefore strictly increasing in $\bar{s}$ on $(\underline{s}, \psi(\underline{s}))$ and constant beyond, so the optimum sits at the corner $\bar{s} = \psi(\underline{s})$.

Hence $p^* = 0$ and $\bar{s}^* = \psi(\underline{s}^*)$ at any Class I optimum. The marginal-seller indifference then pins down

$$e^* = -U^{B,0}(\underline{s}^*; I_S(\underline{s}^*)) < 0,$$

with the strict inequality from $U^{B,0} > 0$ on the data-useful range $\underline{s}^* \in [x, z]$ (cf. Lemma 5; note that for the continuum pool $U^{B,0}(x; [-x, x]) > 0$ by Jensen's inequality, since the pool average strictly exceeds the endpoint x). The designer therefore subsidises sellers.

Step 2 (Class I): Reduction to a one-dimensional optimisation. Once $\underline{s}$ is fixed the implementing instruments $(p, e) = (0, -U^{B,0}(\underline{s}; I_S(\underline{s})))$ are uniquely determined and every buyer's action is Bayesian-rational. Welfare is a function of $\underline{s}$ alone,

$$\widetilde{W}(\underline{s}) := W(\underline{s}, \psi(\underline{s})),$$

and the Class I optimum solves $\max_{\underline{s} \in [x, z]} \widetilde{W}(\underline{s})$. The exponential-family computation in Appendix B confirms that $\widetilde{W}$ admits an interior maximiser $\underline{s}_I^*$, with $\underline{s}_I^*/z \approx 0.89$ at $c/V = 2/3$.

Proof of Class II. We now consider designs with seller pool $I_s^\dagger(\underline{s}) = \{-\underline{s}, +\underline{s}\}$, for $\underline{s} \in (x, z)$ (the feasibility range guaranteed by Theorem 1). Because $U^S = 0$ for any seller in this two-point pool, the S|B margin condition forces

$$p^* = U^{B,0}(\underline{s}; I_s^\dagger(\underline{s})) = MB(\underline{s}, \underline{s}),$$

the expected marginal benefit is

$$MB(s_i, |s|) = \frac{\theta(s_i)f_r(|s|) + (1 - \theta(s_i))f_l(|s|)}{f_r(|s|) + f_l(|s|)} \pi(\tilde{\theta}(s_i, +|s|)) + \frac{\theta(s_i)f_l(|s|) + (1 - \theta(s_i))f_r(|s|)}{f_r(|s|) + f_l(|s|)} \pi(\tilde{\theta}(s_i, -|s|)),$$

the expectation being taken over the sign of s' (equivalently, over ω) at fixed magnitude $|s|$, where $MB(\underline{s}, \underline{s}) > 0$ because $\tilde{\theta}(\underline{s}, \underline{s}) > c/V$ for $\underline{s} \in (x, z)$. Each buyer acquires a signal s' equal to $+\underline{s}$ or $-\underline{s}$ with equal probability (since both seller signals are symmetric), so the conditional distribution degenerates to a two-point distribution on $\{-\underline{s}, +\underline{s}\}$. The investment cutoff $\sigma_h(s)$ for a buyer with signal s is still defined by $\ell(s)\ell(\sigma_h(s)) = h/(1-h)$, and the buyer invests iff her acquired signal exceeds $\sigma_h(s)$.

Since sellers are infinitesimally thin (measure zero), the S|B margin requires that each seller earns zero net payoff; to sustain the seller's indifference between selling and opting out (buying at price p^* yields $U^B(\underline{s}) = U^{B,0}(\underline{s}; I_s^\dagger) - p^* = 0$), the entry fee satisfies $e^* = +\infty$ in the limit (as in the footnote of Section 3.2, by taking the two-point pool as the limit $\epsilon \rightarrow 0$ of a thin continuum $[\underline{s} - \epsilon, \underline{s}]$).

Step 1 (Class II): Characterization of $\bar{s}^*$. With $p^* = MB(\underline{s}, \underline{s})$ fixed by $\underline{s}$, the B|N margin condition determines $\bar{s}$ uniquely. A buyer at threshold $\bar{s}$ is indifferent between buying and not participating:

$$U^{B,0}(\bar{s}; I_s^\dagger(\underline{s})) - p^* = U^N(\bar{s}).$$

Since $U^{B,0}(\bar{s}; I_s^\dagger(\underline{s})) = MB(\bar{s}, \underline{s})$ and $U^N(\bar{s}) = \theta(\bar{s})V - c$ for $\bar{s} \geq z$ (and zero for $\bar{s} < z$), this becomes

$$MB(\bar{s}, \underline{s}) = MB(\underline{s}, \underline{s}) + U^N(\bar{s}).$$

For $\bar{s} \in [z, \psi(\underline{s})]$: $MB(\bar{s}, \underline{s})$ is strictly decreasing in $\bar{s}$ by Lemma 4 (decreasing branch beyond z), while $MB(\underline{s}, \underline{s}) + U^N(\bar{s})$ is non-decreasing. Hence there is at most one solution. Existence follows from the boundary values: at $\bar{s} = \underline{s}$ (degenerate, for reference) the left side exceeds the right, and at $\bar{s} = \psi(\underline{s})$ the left side equals $MB(\underline{s}, \underline{s})$ (the acquired signal is no longer decision-relevant so $MB(\psi(\underline{s}), \underline{s}) = MB(\underline{s}, \underline{s})$) while $U^N(\psi(\underline{s})) > 0$, so the right side exceeds the left; by continuity the unique solution $\bar{s}^*(\underline{s}) \in (z, \psi(\underline{s}))$ exists. In particular $\bar{s}^* < \psi(\underline{s})$, as stated in

Class II (ii).

Step 2 (Class II): Reduction to a one-dimensional optimisation. Given $\underline{s}$, the instruments $(p^*, e^*) = (MB(\underline{s}, \underline{s}), +\infty)$ and the buyer threshold $\bar{s}^*(\underline{s})$ are all determined. Welfare W depends on $(\underline{s}, \bar{s}^*(\underline{s}))$ through the same formulas as in Class I, except that the state-conditional distribution of acquired signals is the two-point measure placing mass $\theta(\underline{s})$ on $+\underline{s}$ and $1 - \theta(\underline{s})$ on $-\underline{s}$ under $\omega = r$ (and symmetrically under $\omega = l$). Define

$$\widehat{W}(\underline{s}) := W(\underline{s}, \bar{s}^*(\underline{s})),$$

and the Class II optimum solves $\max_{\underline{s} \in [x, z]} \widehat{W}(\underline{s})$.

Overall optimum. The learning-optimal design is the class with higher welfare:

$$\max\{\widehat{W}(\underline{s}_I^*), \widehat{W}(\underline{s}_{II}^*)\}.$$

If Class I attains the maximum the design satisfies the properties stated in Class I of the proposition; if Class II attains it the properties of Class II hold. $\square$

Remark 1. Averaging MB against this density recovers the gross buying payoff in Class I exactly,

$$U^{B,0}(\underline{s}^*; I_S(\underline{s}^*)) = \int_0^{\underline{s}^*} MB(\underline{s}^*, s) \frac{f_r(s) + f_r(-s)}{F_r(\underline{s}^*) - F_r(-\underline{s}^*)} ds,$$

the densities enter MB only as the ratio fixing the sign split, and after MB as the level fixing the magnitude frequency, so no probability is double-counted. The Class II evaluation is the degenerate limit of Class I.

A.8 Proof of Theorem 2

Fix $\epsilon > 0$ and consider the candidate profile in which agents with $|s_i| < s_\epsilon$ buy, agents with $|s_i| > s_\epsilon$ sell, and every non-seller takes the optimal Stage-3 action given his own signal and any acquired signal. By symmetry it suffices to treat $s_i > 0$. Let $R_\epsilon^B(s_i)$ denote the expected payoff from buying one signal drawn from $F_\omega(\cdot \mid |s| > s_\epsilon)$ and then acting optimally. Since $s_\epsilon \rightarrow 1$ as $\epsilon \rightarrow 0$ and $\theta(1)V > c$, for all sufficiently small ϵ the seller pool contains signals that are decision-relevant for buyers with positive probability.

Step 1: Seller transfers. Under strict MLRP, the posterior $\tilde{\theta}(s, s_j)$ is strictly increasing in each argument. Thus, after receiving seller signal s_j , the set of buyers who invest R is an upper

interval in their own signal, while the set of buyers who invest L is a lower interval. Hence $A_R(s_j)$ is weakly increasing in s_j and $A_L(s_j)$ is weakly decreasing. The strict cutoff property used in the construction of $\mathcal{T}_\epsilon$ gives $A_R(s_j) \geq A_R(s_\epsilon)$ if and only if $s_j \geq s_\epsilon$, and $A_L(s_j) \geq A_L(-s_\epsilon)$ if and only if $s_j \leq -s_\epsilon$. Consequently a positive seller $s_i > s_\epsilon$ clears the $\omega = r$ gate and fails the $\omega = l$ gate, so he collects the lump-sum reward only in state r , and his expected seller payoff is

$$R_\epsilon^S(s_i) = \theta(s_i)\beta_\epsilon.$$

By symmetry, a negative seller $s_i < -s_\epsilon$ collects the reward only in state l , and the same expression holds with $\theta(|s_i|)$ in place of $\theta(s_i)$.

An agent with $0 < s_i < s_\epsilon$ who deviates to selling clears neither gate: $A_R(s_i) < A_R(s_\epsilon)$ and $A_L(s_i) < A_L(-s_\epsilon)$. Symmetry gives the same conclusion for $-s_\epsilon < s_i < 0$. Thus every low-signal deviation to selling earns zero.

Step 2: A payoff bound for buying. Fix ϵ and $s_i > s_\epsilon$. After any acquired seller signal, let $q \in [0, 1]$ denote the posterior belief that $\omega = r$. The optimal Stage-3 payoff at posterior q is

$$\pi(q) := \max\{qV - c, (1 - q)V - c, 0\}.$$

For every $q \in [0, 1]$, each term in the maximum is bounded above by $V - c$: $qV - c \leq V - c$, $(1 - q)V - c \leq V - c$, and $0 \leq V - c$. Hence $\pi(q) \leq V - c$ pointwise, regardless of whether the optimal action is R , L , or Out . Taking expectations over the acquired signal gives

$$R_\epsilon^B(s_i) \leq V - c.$$

Since $s_i > s_\epsilon$ and θ is increasing, $\theta(s_i) \geq \theta(s_\epsilon) > 0$, so

$$\frac{R_\epsilon^B(s_i)}{\theta(s_i)} \leq \frac{V - c}{\theta(s_\epsilon)}$$

for every $s_i > s_\epsilon$. The same bound holds for $s_i < -s_\epsilon$ by symmetry, replacing $\theta(s_i)$ with $\theta(|s_i|)$.

Step 3: High-signal types sell. For all sufficiently small ϵ , define

$$\beta_\epsilon := \sup_{s_i \in (s_\epsilon, 1)} \frac{R_\epsilon^B(s_i)}{\theta(s_i)}.$$

The previous step implies $\beta_\epsilon \leq (V - c)/\theta(s_\epsilon)$; moreover $\beta_\epsilon > 0$ because $\theta(s_\epsilon)V > c$ for sufficiently small ϵ (recall $s_\epsilon \rightarrow 1$ and $\theta(1)V > c$), so high-signal types have positive standalone

investment value. For every $s_i > s_\epsilon$,

$$R_\epsilon^S(s_i) = \theta(s_i)\beta_\epsilon \geq R_\epsilon^B(s_i),$$

so selling is a best response against buying. The same argument applies to $s_i < -s_\epsilon$ by symmetry. Selling also weakly dominates non-participation under the same calibration, since for $s_i > s_\epsilon$,

$$R_\epsilon^S(s_i) \geq R_\epsilon^B(s_i) \geq \max\{\theta(s_i)V - c, 0\},$$

and the last term is the non-participation payoff. The negative-signal case is symmetric.

Step 4: Low-signal types buy. For $|s_i| < s_\epsilon$, deviating to selling yields zero by Step 1. Buying weakly dominates non-participation because the acquired signal is free and can be ignored. For small ϵ the inequality is strict except for irrelevant indifference cases: if the standalone investment payoff is already positive, the agent can preserve it by ignoring the signal and may improve his decision after observing an extreme countervailing signal; if the standalone payoff is zero, an extreme seller signal induces a profitable investment with positive probability by $\theta(1)V > c$ and continuity. In any case, buying is a best response for every $|s_i| < s_\epsilon$.

The role choices above, together with Bayesian updating and optimal final actions at Stage 3, form a PBE for every $\epsilon \in (0, \bar{\epsilon})$, with $\bar{\epsilon}$ chosen small enough for the preceding inequalities to hold. This proves the claimed role inversion. $\square$

A.9 Proof of Proposition 2

Setup. The seller reward under $\mathcal{T}_\epsilon$ is the flat amount β_ϵ if the relevant correct-action gate is cleared and zero otherwise, with calibration

$$\beta_\epsilon = \sup_{s_i \in (s_\epsilon, 1)} \frac{R_\epsilon^B(s_i)}{\theta(s_i)} \leq \frac{V - c}{\theta(s_\epsilon)}.$$

Thus a positive seller's expected transfer is $\theta(s_i)\beta_\epsilon$, and the symmetric expression applies to a negative seller in state l . This calibration is used only to sustain the role profile. Under Theorem 2, that profile is $I_B = (-s_\epsilon, s_\epsilon)$ and $I_S = (-1, -s_\epsilon) \cup (s_\epsilon, 1)$ with $|I_S| = \epsilon$; only buyers invest. By symmetry $f_r(s) = f_l(-s)$, the welfare under $\omega = r$ and $\omega = l$ coincide, so it suffices to compute $\mathcal{V}(\mathcal{T}_\epsilon)$ conditional on $\omega = r$. We compute welfare by averaging each seller signal's contribution over the buyer pool, showing it is bounded above by $\bar{V}$ but converges to $\bar{V}$ as $\epsilon \rightarrow 0$.

For $s_i \in I_B$ drawing $s_j \sim F_r(\cdot \mid I_S)$: *concordant* sellers ($s_j > s_\epsilon$) trigger R (correct) iff $s_i \geq$

$-\tilde{s}(s_j)$; discordant sellers ($s_j < -s_\epsilon$) trigger L (wrong) iff $s_i \leq \tilde{s}(|s_j|)$. The max / min operations below truncate these regions to the buyer window $(-s_\epsilon, s_\epsilon)$, which limits welfare relative to the unrestricted benchmark. Define the per-seller-signal welfare contribution

$$V_w(s; \epsilon) := \theta(s)[F_r(s_\epsilon) - F_r(\max\{-s_\epsilon, -\tilde{s}(s)\})](V - c) - (1 - \theta(s))[F_r(\min\{s_\epsilon, \tilde{s}(s)\}) - F_r(-s_\epsilon)]c.$$

Averaging over s_j and using $f_r(s)/(f_r(s) + f_l(s)) = \theta(s)$, $f_l(s)/(f_r(s) + f_l(s)) = (1 - \theta(s))$ yields

$$\mathcal{V}(\mathcal{T}_\epsilon) = \int_{s_\epsilon}^1 V_w(s_j; \epsilon) \frac{f_r(s_j) + f_l(s_j)}{\epsilon} ds_j,$$

a weighted average of $V_w(\cdot; \epsilon)$ on $(s_\epsilon, 1)$ since $\int_{s_\epsilon}^1 (f_r(s) + f_l(s))/\epsilon ds = 1$ by symmetry. This integral contains no factor involving β_ϵ , $A_R(s_j)$, or $A_L(s_j)$; it uses only the seller-signal distribution induced by I_S and the buyers' optimal final actions. Hence no step of the welfare calculation assumes the old proportional reward structure.

Upper bound $\mathcal{V} \leq \bar{V}$. Define

$$W(s) := \theta(s)(F_r(s) - F_r(-\tilde{s}(s)))(V - c) - (1 - \theta(s))(F_r(\tilde{s}(s)) - F_r(-s))c,$$

so $\bar{V} = \lim_{s \rightarrow 1} W(s)$. W is increasing in s (more-informative seller signals are Blackwell-better). Crucially, $V_w(s_j; \epsilon)$ restricts buyers to $(-s_\epsilon, s_\epsilon)$ while $W(s_j)$ allows all agents in $(-s_j, s_j)$ to react; this restriction reduces welfare. The omitted buyers contribute non-negative net welfare because on the seller range $\theta(s_j) > c/V$ and MLRP imply $\theta(s_j)(V - c) \int f_r \geq (1 - \theta(s_j))c \int f_l$ over (s_ϵ, s_j) , so $V_w(s_j; \epsilon) \leq W(s_j) \leq \bar{V}$. Averaging gives $\mathcal{V} \leq \bar{V}$.

Convergence $\mathcal{V} \rightarrow \bar{V}$. It suffices to show $V_w(s_j; \epsilon) \rightarrow \bar{V}$ uniformly on $(s_\epsilon, 1)$. As $\epsilon \rightarrow 0$, the buyer window $(-s_\epsilon, s_\epsilon)$ expands toward $(-1, 1)$ and seller signals concentrate near ± 1 , so the restriction imposed by the buyer window becomes negligible.

Case $c/V > 1/2$. With $\tilde{s}_0 := \lim_{s \rightarrow 1} \tilde{s}(s) < 1$, for small ϵ we have $\tilde{s}(s_j) \leq \tilde{s}_0 < s_\epsilon$ on $(s_\epsilon, 1)$, so the inner max / min become $-\tilde{s}(s_j), \tilde{s}(s_j)$. As $s_\epsilon, s_j \rightarrow 1$, the buyer window contains all relevant buyers (those who react to signal s_j), so $V_w(s_j; \epsilon) \rightarrow W(s_j) \rightarrow \bar{V}$ uniformly.

Case $c/V \leq 1/2$. $\tilde{s}(s_j) = s_j > s_\epsilon$, so both bracketed factors equal $1 - \epsilon$ and $V_w(s_j; \epsilon) = (1 - \epsilon)[\theta(s_j)V - c] \rightarrow \theta(1)V - c = \bar{V}$ uniformly.

Combined with the upper bound, $\lim_{\epsilon \downarrow 0} \mathcal{V}(\mathcal{T}_\epsilon) = \bar{V}$ and therefore $\sup_{\epsilon > 0} \mathcal{V}(\mathcal{T}_\epsilon) = \bar{V}$. $\square$

A.10 Proof of Proposition 3

Setup. Let $\Pi(\epsilon)$ denote the supremum of designer profit over $\mathcal{T} \in \mathbb{T}(\epsilon) := \{\mathcal{V}(\mathcal{T}) \geq \bar{V} - \epsilon\}$. Case 1 shows $\Pi(\epsilon) \rightarrow 0$ when beliefs are unbounded or $c/V \leq 1/2$; Case 2 constructs a design with profit $\rightarrow \theta(1)(1 - \theta(1))(2c - V) > 0$ when beliefs are bounded and $c/V > 1/2$.

Case 1: $\Pi(\epsilon) \rightarrow 0$. Fix $\mathcal{T} \in \mathbb{T}(\epsilon)$. Buyer participation gives $-T_B \leq U^{B,0}(s_i; I_S) - U^N(s_i)$ for $s_i \in I_B$, where $T_B := T_B(|I_S|, |I_B|)$ is constant across all buyers. The IC then forces

$$-T_B \leq \inf_{s_i \in I_B} [U^{B,0}(s_i; I_S) - U^N(s_i)] := \kappa^*,$$

attained at the marginal buyer $|s_i| \rightarrow s_\epsilon$ by Lemma 4. With seller-side surplus bounded by $(V - c)|I_S| = (V - c)\epsilon \rightarrow 0$, $\Delta(\mathcal{T}) \leq -T_B|I_B| + (V - c)\epsilon \leq \kappa^*|I_B| + (V - c)\epsilon$, and it suffices to show the infimum vanishes.

Subcase 1a (unbounded, $\theta(1) = 1$). As $|s_i| \rightarrow s_\epsilon \rightarrow 1$, $\theta(s_i) \rightarrow 1$, so $\tilde{\theta}(s_i, s_j) \rightarrow 1$ for every s_j : every buyer invests in $\text{sgn}(s_j)$ (action-independence) and $U^{B,0}(s_\epsilon; I_S) \rightarrow V - c = U^N$ (vanishing surplus).

Subcase 1b ($c/V \leq 1/2$). The *Out*-optimal region $\tilde{\theta} \in [1 - c/V, c/V]$ is empty, so buyers always invest. Since $|s_j| > |s_i|$, MLRP gives $\text{sgn}(\ell(s_i)\ell(s_j)) = \text{sgn}(s_j)$, so the action depends only on s_j (action-independence). Summing over the four $(\omega, s_j = \pm 1)$ cells, the seller-signal terms cancel and $U^{B,0}(s_i; I_S) \rightarrow \theta(s_i)V - c = U^N(s_i)$ at every s_i (vanishing surplus).

In both subcases $\kappa^* \rightarrow 0$, giving $\Pi(\epsilon) \rightarrow 0$.

Case 2: $\Pi(\epsilon) \rightarrow \theta(1)(1 - \theta(1))(2c - V) > 0$. Augment $\mathcal{T}_\epsilon$ from Theorem 2 with a uniform participation fee $\kappa' > 0$ on every market participant. Lump-sum fees do not distort final actions, and they preserve the role profile whenever every buyer and seller has surplus at least κ' . Thus, once feasibility is verified below, $\mathcal{V}(\mathcal{T}_{\epsilon, \kappa'}) = \mathcal{V}(\mathcal{T}_\epsilon) \rightarrow \bar{V}$.

Limiting marginal-buyer surplus. Sellers concentrate near $|s| = 1$. At $|s_i| = s_\epsilon$, summing over the four $(\omega, s_j = \pm 1)$ cells with weights $(\theta(1)^2, \theta(1)(1 - \theta(1)), (1 - \theta(1))^2, \theta(1)(1 - \theta(1)))$:

- Concordant cells $(s_i, s_j \text{ aligned with } \omega)$: $\tilde{\theta} \rightarrow 1$, invest correctly $\Rightarrow V - c$ (prob $\theta(1)^2$) or $-c$ when s_i, s_j aligned but ω opposite (prob $(1 - \theta(1))^2$).
- Discordant cells: $\tilde{\theta}(1, -1) = 1/2 < c/V$, so *Out*, payoff 0.

Hence $U^{B,0}(s_\epsilon; I_S) \rightarrow \theta(1)^2(V - c) - (1 - \theta(1))^2c$ while $U^N(s_\epsilon) \rightarrow \theta(1)V - c$, yielding

$$\kappa := \lim[U^{B,0} - U^N] = \theta(1)(1 - \theta(1))(2c - V) > 0,$$

strictly positive since $\theta(1) < 1$ and $c/V > 1/2$.

Feasibility of $\kappa' < \kappa$. For buyers, $U^{B,0} - U^N$ is decreasing in $|s_i|$ (Lemma 4), so the marginal buyer binds and $\kappa' < \kappa$ is feasible. For sellers, since $\beta_\epsilon \geq R_\epsilon^B(1)/\theta(1)$, each seller's surplus satisfies

$$R_\epsilon^S(s_i) - U^N(s_i) = \theta(s_i)\beta_\epsilon - U^N(s_i) \geq \theta(s_i)\frac{R_\epsilon^B(1)}{\theta(1)} - (\theta(s_i)V - c) \longrightarrow \kappa$$

as $\epsilon \rightarrow 0$, because sellers concentrate in a vanishing neighborhood of $|s| = 1$. The marginal buyer binds at exactly κ by Lemma 4, so the minimum participant surplus converges to κ , and $\kappa' < \kappa$ is feasible for all sufficiently small ϵ .

Profit. The designer collects κ' from unit measure of participants and pays only the gated seller rewards. Their total expected amount is at most $\beta_\epsilon \epsilon = O(\epsilon)$, because $|I_S| = \epsilon$ and $\beta_\epsilon \leq (V - c)/\theta(s_\epsilon) = O(1)$ since $\theta(s_\epsilon) > 1/2$. Therefore $\Delta(\mathcal{T}_{\epsilon, \kappa'}) = \kappa' - O(\epsilon)$. Letting $\kappa' \uparrow \kappa$, $\epsilon \downarrow 0$ gives $\lim \Pi(\epsilon) \geq \kappa$. $\square$

A.11 Proof of Proposition 4

State-dependent markets: $\sup_{\mathcal{T} \in \mathbb{T}_D} \mathcal{V} = \bar{\mathcal{V}}$. Adapt $\mathcal{T}_\epsilon$ from Theorem 2: keep the same gated flat reward and charge each seller an entry fee $e_\epsilon := \theta(s_\epsilon)\beta_\epsilon > 0$. A seller with $|s_i| < s_\epsilon$ clears no gate and earns zero before the fee, while a seller with $|s_i| > s_\epsilon$ clears the gate in the state indicated by his signal and earns expected reward $\theta(|s_i|)\beta_\epsilon > e_\epsilon$. Hence agents with $|s_i| > s_\epsilon$ strictly prefer selling and those with $|s_i| < s_\epsilon$ do not. Sellers may additionally buy in the dual-role model; access to an extra signal weakly raises $\mathcal{V}$. The seller pool has mass $\epsilon \rightarrow 0$, so by Proposition 2, $\sup_{\mathbb{T}_D} \mathcal{V} = \bar{\mathcal{V}}$.

State-independent markets: $\sup_{\mathcal{T} \in \mathbb{T}_I} \mathcal{V} = \bar{\mathcal{V}}$. Since selling is costless in the dual-role model (sellers invest in Stage 3 regardless), at $\hat{T}_S = 0, e = 0$ every agent is indifferent between the role bundle $\{\mathbf{S}, \mathbf{B}\}$ and the pure buyer role $\mathbf{B}$, yielding $U^S = 0$. By indifference breaking, designate agents with $|s_i| > 1 - \epsilon$ as sellers. Buyers acquire signals drawn from this near-degenerate distribution; as $\epsilon \rightarrow 0$, acquired signals concentrate near ± 1 and $\mathcal{V} \rightarrow \bar{\mathcal{V}}$ by the same welfare calculation as in Proposition 2. Hence $\sup_{\mathbb{T}_I} \mathcal{V} = \bar{\mathcal{V}}$.

Necessity of (i): zero seller revenue. Suppose $\hat{T}_S > 0$. In the dual-role model, selling is costless: a seller invests in Stage 3 regardless of their role, so selling adds exactly $\hat{T}_S > 0$ to an agent's payoff at zero opportunity cost. Every agent therefore *strictly* prefers the role bundle $\{\mathbf{S}, \mathbf{B}\}$ over pure buyer, with no indifference to break. Hence $I_S = (-1, 1)$ in every equilibrium with trade, and each buyer draws an acquired signal from the unconditional distribution $F_\omega(\cdot)$.

The resulting information structure—each agent observes two i.i.d. draws (s_i, s_j) from F_ω —is strictly Blackwell-dominated by a degenerate distribution concentrating seller signals at ± 1 : the latter reveals the state perfectly as the seller signal approaches ± 1 , yielding welfare $\rightarrow \bar{V}$, whereas i.i.d. draws from the interior of $(-1, 1)$ leave positive probability mass on incorrect investment even in the limit. Hence $\mathcal{V}(\mathcal{T}) < \bar{V}$, proving necessity of (i) by contrapositive.

Necessity of (ii): indifference must be broken in favour of the most-informed. By (i), any mechanism approaching $\bar{V}$ must have $\hat{T}_S = 0$. With $\hat{T}_S = 0$, selling adds nothing to an agent's investment return and carries no cost, so *every* agent is indifferent between selling and not selling. The equilibrium seller pool is therefore entirely determined by whatever indifference-breaking rule is in force; no type has a strict preference that pins down participation. If the indifference-breaking rule selects sellers with $|s_i| \leq M$ for some $M < 1$ (i.e. every seller signal satisfies $|s'| \leq M$), then $W(\cdot)$ is bounded above by $W(M) < \bar{V}$, and the welfare integral gives $\mathcal{V} \leq W(M) < \bar{V}$. Hence approaching $\bar{V}$ requires the indifference-breaking rule to place selling on agents with $|s_i| \rightarrow 1$, i.e. only the most-informed agents sell. $\square$

B Supplementary Materials

B.1 Equilibrium Uniqueness and Comparative Statics

We give a sufficient condition under which the MB–LS equilibrium is unique for every (p, e) and derive comparative statics. The key step is monotonicity of the two margin functions, building on Lemma 4 from Appendix A.3.

Lemma 7 (Monotonicity of Margin Functions). *Both $M(\underline{s}, \bar{s})$ and $m(\underline{s}, \bar{s})$ are strictly increasing in $\underline{s}$ and strictly decreasing in $\bar{s}$.*

Proof. Write $\sigma(s)$ for the buyer signal threshold solving $\tilde{\theta}(s, \sigma(s)) = c/V$ and $F_w(s_1, s_2) := F_w(s_1) - F_w(s_2)$.

(a) M . Using symmetry $|I_s| = F_r(s, -\underline{s}) = F_l(\underline{s}, -\underline{s})$, write $M = Q(\underline{s})/[F_w(\bar{s}, -\bar{s})]$ with

$$Q(\underline{s}) := U^{B,0}(\underline{s}; I_s(\underline{s})) \cdot |I_s(\underline{s})| = \theta(\underline{s})F_r(\underline{s}, \sigma(\underline{s}))(V - c) - [1 - \theta(\underline{s})]F_l(\underline{s}, \sigma(\underline{s}))c.$$

The terms in $\sigma'(\underline{s})$ cancel by the definition of σ , leaving

$$Q'(\underline{s}) = \theta'(\underline{s})[F_r(V - c) + F_l c] + [\theta(\underline{s})f_r(\underline{s})(V - c) - (1 - \theta(\underline{s}))f_l(\underline{s})c].$$

The first bracket is positive ($\theta' > 0$ by MLRP); strict MLRP at $\underline{s} > \sigma(\underline{s})$ gives $f_r(\underline{s})/f_l(\underline{s}) > (1 - \theta(\underline{s}))c/[\theta(\underline{s})(V - c)]$, making the second bracket positive too. Hence $Q' > 0$, so M rises in $\underline{s}$ (numerator only) and falls in $\bar{s}$ (denominator only).

(b) m in $\bar{s}$. Direct differentiation of $m = U^{B,0}(\bar{s}; I_s) - U^N(\bar{s})$ gives

$$m_{\bar{s}} = -\theta'(\bar{s})[\alpha(\bar{s}; I_s)V + (1 - \alpha - \beta)c] < 0,$$

since the bracket is positive whenever $\bar{s} < \psi(\underline{s})$ (data is informative). This also follows from Lemma 4.

(c) m in $\underline{s}$. Expanding seller pool always lowers Type II error ($\partial\beta/\partial\underline{s} < 0$); $\partial\alpha/\partial\underline{s}$ has ambiguous sign. After cancelling using $\sigma(\bar{s})$, $m_{\underline{s}} > 0$ reduces to

$$\frac{f_l(\sigma(\bar{s}))}{f_r(\sigma(\bar{s}))} \left(-\frac{\partial\alpha}{\partial\underline{s}} \right) - \frac{\partial\beta}{\partial\underline{s}} > 0.$$

If $\partial\alpha/\partial\underline{s} \leq 0$ both terms are non-negative and we are done. Otherwise, MLRP gives $f_l(-\underline{s})/f_r(-\underline{s}) > f_l(\sigma(\bar{s}))/f_r(\sigma(\bar{s}))$ (since $-\underline{s} < \sigma(\bar{s})$), so it suffices to verify the inequality with the larger ratio. Setting $(\alpha_s, \beta_s) := (\partial\alpha/\partial\underline{s}, \partial\beta/\partial\underline{s}) \cdot |I_s|^2$ and computing,

$$\alpha_s - \beta_s = [f_r(\underline{s}) + f_r(-\underline{s})] \cdot F_r(-\sigma(\bar{s}), \sigma(\bar{s})) > 0,$$

and substitution reduces the claim to $f_r(\underline{s})/f_r(-\underline{s}) > F_r(\underline{s}, -\sigma(\bar{s}))/F_r(\sigma(\bar{s}), -\underline{s})$, which holds by strict MLRP. $\square$

Uniqueness under $\varphi'(\underline{s}) < 0$. Along the equilibrium path $\bar{s} = \varphi(\underline{s})$, define $\mathcal{P}(\underline{s}) := m(\underline{s}, \varphi(\underline{s})) = M(\underline{s}, \varphi(\underline{s})) = p$. By Lemma 7, $m_{\underline{s}} > 0$ and $m_{\bar{s}} < 0$, so

$$\mathcal{P}'(\underline{s}) = m_{\underline{s}} + m_{\bar{s}} \cdot \varphi'(\underline{s}) > 0 \quad \text{whenever } \varphi'(\underline{s}) < 0.$$

Strict monotonicity of $\mathcal{P}$ rules out two distinct $\underline{s}$ supporting the same p , giving uniqueness; existence is Theorem 1. $\square$

Comparative statics. Since $\mathcal{P}' > 0$ and $\varphi' < 0$, raising p increases $\underline{s}$ and lowers $\bar{s}$: both thresholds move toward z and the market shrinks. A fee e enters M additively, so restoring $M = p$ after $e \uparrow$ requires lower $\underline{s}$ (as $M_{\underline{s}} > 0$) and hence lower $\bar{s}$: both thresholds shift toward 0. A subsidy ($e < 0$) reverses the signs. $\square$

B.2 Example: Exponential Family of Private Signals

We specialize in the exponential signal family $f_r(s) = ke^{s+a}$, $f_l(s) = ke^{-s+a}$ in $(-1, 1)$, with the location parameter $a = -1$ and the normalization constant $k = e^2/(e^2 - 1)$, and verify all major results of the paper in closed form. All numerical evaluations use $c/V = 2/3$ (so $h = 2/3$, $z = \frac{\ln 2}{2} \approx 0.347$, and $V = 1$) as the running example. The accompanying scripts `simulate_payoffs.py` and `simulate_dynamics.py` reproduce every computation below.

Primitives

The posterior is $\theta(s) = e^{2s}/(e^{2s} + 1)$. The break-even signal satisfies $\theta(z) = h$, giving $z = \frac{1}{2} \ln \frac{h}{1-h}$. The decision-inversion threshold is additive: $\sigma_h(s) = z - s$, and the data-relevance frontier is $\psi(\underline{s}) = \underline{s} + z$, yielding a constant-width buyer window. The feasibility lower bound for $\underline{s}$ is $x = z/2$. The conditional CDF of the seller pool $I_s = [-\underline{s}, \underline{s}]$ is

$$G(s'; \underline{s}) = \frac{e^{s'} - e^{-\underline{s}}}{e^{\underline{s}} - e^{-\underline{s}}} = \frac{e^{s'} - e^{-\underline{s}}}{2 \sinh \underline{s}},$$

and the Type I/II errors specialise to

$$\alpha(s; I_s) = \frac{e^{z-s} - e^{-\underline{s}}}{2 \sinh \underline{s}}, \quad \beta(s; I_s) = \frac{e^{s-z} - e^{-\underline{s}}}{2 \sinh \underline{s}}, \quad s \in [\underline{s}, \underline{s} + z].$$

The agents have bounded private beliefs: $\theta(1) = e^2/(e^2 + 1) \approx 0.881 < 1$.

State-Independent Equilibrium for Arbitrary (p, e)

The seller-buyer margin condition $M(\underline{s}, \bar{s}) = p$ (equation (1) in Theorem 1) and the buyer-investor margin condition $m(\underline{s}, \bar{s}) = p$ (equation (2)) specialise to

$$M(\underline{s}, \bar{s}) = \frac{2 \sinh \underline{s} \cdot U^{B,0}(\underline{s}; I_s)}{2 \sinh \bar{s}} = p,$$

$$m(\underline{s}, \bar{s}) = U^{B,0}(\bar{s}; I_s) - U^N(\bar{s}) = p,$$

where $U^{B,0}(s; I_s) = \theta(s)[1 - \alpha(s)](V - c) - [1 - \theta(s)]\beta(s)c$ and $U^N(s) = \theta(s)V - c$. The equilibrium path satisfies $\partial \bar{s} / \partial \underline{s} < 0$ (verified for exponential distributions by direct computation of $LHS < 0$ and $RHS > 0$ in the implicit-function-theorem decomposition of Section B.1), confirming a unique threshold pair for each admissible (p, e) .

Learning-Optimal State-Independent Design (Proposition 1)

Proposition 1 reduces the welfare problem $W = (V-c)P_R - cP_L$ to the better of two candidates.

Class I. At the corner $p^* = 0$, $\bar{s}^* = \underline{s}^* + z$ (Example I), the FOC reduces to the cubic in $\xi = e^{2\underline{s}}$, $\Lambda = z e^z$:

$$\xi^3 - (3 + 2\Lambda)\xi^2 + (3e^{2z} - 2\Lambda)\xi - e^{2z} = 0,$$

with unique root $\xi^* \in (e^z, e^{2z})$ given by Cardano's trigonometric form

$$\xi^* = \frac{3+2\Lambda}{3} + 2\sqrt{-\frac{p}{3}} \cos\left(\frac{1}{3} \arccos\left(\frac{3q}{2p} \sqrt{-\frac{3}{p}}\right)\right), \quad (4)$$

$p = 3e^{2z} - 2\Lambda - (3 + 2\Lambda)^2/3$, $q = -2(3 + 2\Lambda)^3/27 + (3 + 2\Lambda)(3e^{2z} - 2\Lambda)/3 - e^{2z}$. Then $\underline{s}^* = \frac{1}{2} \ln \xi^*$ and $|e^*| = U^{B,0}(\underline{s}^*, I_S^*)$.

Class II. Use the closed forms $p^* = (V - c)(A - \eta)/(1 + a)^2$ and $\xi^* = [\eta(1 + a + A) - A]/[1 + a + A - \eta]$ from Example II ($a = e^{2\underline{s}}$, $A = a^2$), and maximise

$$\hat{V}(\underline{s}) = \int_{\underline{s} < |s| < \bar{s}^*(\underline{s})} [\theta(s)\theta(\underline{s})(V-c) - (1-\theta(s))(1-\theta(\underline{s}))c] \frac{f_r + f_l}{2} ds + \int_{|s| > \bar{s}^*(\underline{s})} [\theta(s)V - c] \frac{f_r + f_l}{2} ds$$

numerically over $\underline{s} \in [x, z]$ (the buyer integrand posterior-weights the two-point seller signal $\pm \underline{s}$).

Comparison. Class II welfare-dominates class I throughout the examined range (Tables 3, 2): at $c/V = 2/3$, $W_{II}/V \approx 0.0964 > W_I/V \approx 0.0928$, with the gap widening in c/V . Class II uses a smaller but boundary-concentrated seller pool ($\underline{s}^*/z \approx 0.65$ vs. 0.89), a strictly interior $\bar{s}^* < \psi(\underline{s}^*)$, and replaces the class I subsidy with positive data revenue. The ranking relies on the indifference-breaking selection $|s_i| = \underline{s}^*$; absent it, the designer falls back to class I.

Learning-Optimal State-Dependent Design (Propositions 2 and 3)

Under the gated state-dependent mechanism $\mathcal{T}_\epsilon$, the corrected seller transfer affects implementation but not the induced information allocation: sellers still concentrate near ± 1 as $\epsilon \rightarrow 0$, and buyers in the central region receive those signals for free. The payoff simulation therefore plots the seller payoff as zero below s_ϵ and as $\theta(|s|)\beta_\epsilon$ once the correct-action gate clears, rather than scaling the transfer by the realised correct-action rate. The welfare calculation is unchanged because it depends only on the role allocation and the buyers' induced decisions.

c/V	$\underline{s}^*/z$	$\bar{s}^*/z$	p^*/V	P_R	P_L	W/V (W/V class I)
0.55	0.665	1.332	0.0082	0.687	0.228	0.184 (0.184)
0.60	0.661	1.330	0.0157	0.639	0.189	0.143 (0.141)
0.65	0.654	1.326	0.0219	0.587	0.152	0.107 (0.104)
2/3	0.652	1.324	0.0236	0.569	0.140	0.096 (0.093)
0.70	0.645	1.321	0.0262	0.529	0.116	0.078 (0.073)
0.75	0.632	1.316	0.0280	0.463	0.081	0.055 (0.048)

Table 2: Learning-optimal class II design for the exponential family; class I welfare in parentheses. Class II strictly dominates throughout, increasingly so as c/V rises.

In the limit, a buyer sees $s' > 0$ w.p. $\theta(1)$ and invests correctly iff $s_i \geq z - 1$; with $s' < 0$, she would need $s_i > 1$. Hence

$$P_R = \theta(1) \cdot \frac{e^2 - e^z}{e^2 - 1} \approx 0.824, \quad P_L = [1 - \theta(1)] F_r(1 - z) \approx 0.079,$$

so $\bar{V} = P_R(V - c) - P_L \cdot c \approx 0.222V$ —over double the state-independent optimum. Bounded beliefs and $c/V > 1/2$ give a per-capita profit limit $\kappa = \theta(1)(1 - \theta(1))(2c - V) \approx 0.035V$ (Proposition 3).

Dynamic State-Independent Markets (Section 5.1)

Across rounds we track each agent’s *belief* $b_i^{(\tau)} \in \mathbb{R}$, defined as the posterior log-likelihood ratio for $\omega = r$ versus $\omega = l$ given everything agent i has observed by round τ . Beliefs are a sufficient statistic for action choice via $\theta(b_i^{(\tau)}) = 1/(1 + e^{-2b_i^{(\tau)}}) = \Pr(\omega = r \mid \text{info}_i^{(\tau)})$; the designer’s threshold rule and pool boundaries depend only on $|b|$, so the *design* is state-symmetric. For the exponential family the initial belief is $b_i^{(0)} = s_i$, but the round-1 empirical distribution of $\{b_i^{(0)}\}$ is *not* symmetric under $\omega = r$: it is the skewed density f_r , so positive beliefs are over-represented from round 1 onward even though the designer treats $\pm b$ symmetrically. In every subsequent round a buyer who is matched to seller j updates additively, $b_i^{(\tau)} = b_i^{(\tau-1)} + b_j^{(\tau-1)}$, while non-buyers carry their belief unchanged. The seller pool, buyer pool, and non-participant set in round τ are therefore $\{i : |b_i^{(\tau-1)}| < \underline{s}^{*,(\tau)}\}$, $\{i : \underline{s}^{*,(\tau)} \leq |b_i^{(\tau-1)}| < \bar{s}^{*,(\tau)}\}$, and $\{i : |b_i^{(\tau-1)}| \geq \bar{s}^{*,(\tau)}\}$ respectively.

We simulate the class I learning-optimal market at $N = 50,000$, $c/V = 2/3$, $\omega = r$, averaging round-by-round statistics over five independent seeds. Each round the designer re-solves Proposition 1 at the current empirical belief distribution $\{b_i^{(\tau-1)}\}_{i=1}^N$. In round 1 the designer admits $\approx 25\%$ of the population as sellers and $\approx 33\%$ as buyers; the remaining $\approx 42\%$ are non-participants who invest on their private signal alone. Sellers cluster near low $|b|$ where beliefs

are weak, so the designer pays a per-capita subsidy of $\approx 0.007V$ to keep marginally-informed agents on the sell side rather than the buy side. With P_R (resp. P_L) denoting the population fraction making the correct (resp. incorrect) investment, aggregate welfare $W = (V-c)P_R - cP_L$ rises from the no-trade benchmark $0.088V$ to $W^{(1)} \approx 0.092V$. The gain comes almost entirely from a drop in P_L ($0.146 \rightarrow 0.128$); P_R barely moves.

Over rounds 2–12 the market contracts gradually rather than collapsing. The buyer pool thins fastest, since each active buyer receives a fresh additive shock to b and a substantial fraction is pushed past the upper edge $\bar{s}^{*,(\tau)}$ and exits to non-participation; by round 12 the buyer count has fallen from $\approx 12,000$ to $\approx 2,600$. Welfare drifts up to $W^{(12)}/V \approx 0.103$ while the designer's cumulative subsidy reaches $0.038V$ per capita. Twelve rounds of trade thus deliver $\approx 0.015V$ of cumulative welfare gain—only modestly more than round 1's $0.004V$ —confirming the main-text mechanism: the designer is unwilling to enlist more-informed agents as sellers (which would shrink the investing base), so traded information remains weak and gains accumulate slowly at growing fiscal cost.

Class II. The class-II analogue keeps the same belief-array state $\{b_i^{(\tau)}\}$ but treats the degenerate two-point seller pool $\{\pm \underline{s}^{*,(\tau)}\}$ as a *signaling channel*, not as a subset of the population, consistent with the static derivation in Table 2. For each candidate $\underline{s} \in [z/2, z]$ the designer computes the interior $\bar{s}^*(\underline{s})$ and price $p^*(\underline{s})$ from the static closed forms (pointwise indifference at the marginal seller and the marginal buyer, hence family- and round-independent). Buyers $\{i : \underline{s} \leq |b_i^{(\tau-1)}| < \bar{s}^*(\underline{s})\}$ receive a Bernoulli shock $\varepsilon_i \in \{-1, +1\}$ with $\Pr(\varepsilon_i = +1) = \theta(\underline{s})$ under $\omega = r$ and update $b_i^{(\tau)} = b_i^{(\tau-1)} + \varepsilon_i \underline{s}$; the indifference cohort $\{i : |b_i^{(\tau-1)}| < \underline{s}\}$ pays $p^{*,(\tau)}$ but does not update beliefs; non-participants do neither. The designer maximises realised welfare $W^{(\tau)}(\underline{s}) = (V-c)P_R^{(\tau)} - cP_L^{(\tau)}$ over $[z/2, z]$ subject to $p^*(\underline{s}) > 0$, $|B^{(\tau)}| > 0$, and $W^{(\tau)} \geq W_{NT}^{(\tau)}$. The investment threshold z and the feasibility window $[z/2, z]$ are invariants of c/V and the posterior identity $\theta(b) = 1/(1 + e^{-2b})$; only the empirical distribution of beliefs inside that window shifts across rounds.

Iterating the class-II design at the same $(N, c/V, \omega)$, round 1 reproduces the static class-II optimum of Table 2: $\underline{s}^{*,(1)}/z \approx 0.65$, $\bar{s}^{*,(1)}/z \approx 1.32$, $p^{*,(1)} \approx 0.024V$, $W^{(1)} \approx 0.096V$ (vs. no-trade $0.088V$), with $\sim 19\%$ of agents in the indifference cohort and $\sim 21\%$ buying. The designer now *collects* revenue $p^{*,(1)}|B^{(1)}|/N \approx 0.005V$ rather than paying a subsidy. From round 2 onward the buyer pool shrinks faster than under class I: each round's $\pm \underline{s}^{*,(\tau)}$ shock pushes a substantial fraction of buyers across $\bar{s}^{*,(\tau)}$, and the requirement $p^* > 0$ pins $\underline{s}^{*,(\tau)}$ strictly above $z/2$, preventing the designer from lowering the bar to retain marginal buyers. Trade terminates within 4–6 rounds (cf. the 12-round contraction under class I), cumulative welfare reaches $W^\infty \approx 0.099V$ vs. $0.103V$ for class I, and the designer's cumulative revenue

is $\approx 0.010V$ per capita—a sign reversal relative to class I’s $-0.038V$ subsidy. Class II thus dominates class I in budget terms but converges to slightly lower long-run welfare, illustrating the mechanism in Section 5.1: positive p^* accelerates the exit of agents who become sufficiently informed.

Summary Comparison at $c/V = 2/3$

	Learn-opt I	Learn-opt II	State-dep (gated)	No trade
$\underline{s}^*/z$	0.893	0.652	—	—
$\bar{s}^*/z$	1.893	1.324	—	—
$(p^*, e^*)/V$	$(0, -0.027)$	$(+0.024, +\infty)$	—	—
P_R / P_L	0.525/0.123	0.569/0.140	0.824/0.079	0.555/0.144
W/V	0.0928	0.0964	0.222	0.089

Class II welfare-dominates among state-independent designs (the $e^* \rightarrow +\infty$ entry is the per-seller fee in the vanishing-mass limit; total transfers stay finite). The gated state-dependent design dominates all: P_R jumps to 0.824, P_L halves, and the participation-fee variant yields the profit limit above. The transfer revision changes the finite- ϵ seller-payoff calibration, not these P_R/P_L welfare entries. The $W/V \approx 0.222$ ceiling vs. first-best $1/3$ reflects the $P_L \approx 0.079$ floor from bounded beliefs ($\theta(1) \approx 0.881$).

Table 3 reports the class I learning-optimal design across the feasible range of c/V ; Table 2 reports the class II learning-optimum in parallel.

c/V	$\underline{s}^*/z$	e^*/V	W/V
0.55	0.896	−0.0087	0.184
0.60	0.895	−0.0170	0.141
0.65	0.894	−0.0245	0.104
2/3	0.893	−0.0267	0.093
0.70	0.892	−0.0307	0.073
0.75	0.889	−0.0353	0.048

Table 3: Class I learning-optimal (W) state-independent design across c/V , exponential signal family. Sign convention: $e^* < 0$ is a subsidy. The class II learning-optimum (which dominates class I throughout, see Table 2) is reported separately. Values computed from `simulate_payoffs.py` (Block 3).